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Monitoring mangrove forest change and its impacts on the environment

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ABSTRACT

With a focus on the Sundarbans, this study uses remote sensing methods to thoroughly evaluate the dynamics of mangrove forest change and the ensuing environmental effects from 2004 to 2022. Through a thorough time-series analysis, the research reveals the temporal fluctuations in forest cover by utilizing Vegetation Indices and Land Use and Land Cover analysis. The deep significance of these shifts in forest cover is illuminated by the association with specific environmental conditions. The study's conclusions show a concerning trend in the Sundarbans Forest Cover, with a steady reduction occurring at an average annual rate of 2.66%. The study period's sustained decline in mangrove cover highlights the continuous and growing danger to this vital ecosystem. Concurrently, the study notes a concerning increase in urban areas, mostly linked to fishing and sawmills. This indicates that human activity is a major factor in the destruction of the Sundarbans, highlighting the pressing need for focused conservation efforts. The study uses forecasts and simulations to anticipate trends in changes in forest cover up to 2030, extending its investigation beyond the observed timeframe. These forecasts demonstrate geographical differences in the distribution of these changes in addition to confirming the continuance of the downward trend. The need for proactive conservation initiatives is highlighted by the simulations' ability to provide vital foresight in addressing the long-term implications of mangrove destruction. The relationship between changes in forest cover and the ecosystem as a whole is illustrated by correlation analysis with a few chosen environmental indicators, such as precipitation, soil moisture, land surface temperature (LST), and CO2 emissions. This association provides important information for comprehending the wider ecological ramifications of these changes by highlighting the extensive effects of mangrove destruction on important environmental variables. Lagged correlation analysis, which explores the temporal dynamics of the relationship between forest changes and environmental data, is a noteworthy feature of the study. This analysis shows that the effects of deforestation eventually overwhelm the ecosystem, even in the face of initial resistance. The lagged correlation highlights the necessity of taking time lags into account in order to fully comprehend the range of environmental repercussions, as well as the delayed but significant effects of mangrove deforestation on the surrounding ecology. Both mangrove forests and the larger planetary environment will suffer from the predicted trends this study presents. A cycle of diminishing vegetation vigour is set off by the continuous loss of forest cover, which exacerbates environmental harm. This study's methodology is useful for evaluating and tracking ecosystem changes because it can be applied to many kinds of forests. The researchers do concede that on-site observations could yield better results, noting that direct, ground-level evaluations should be conducted in addition to remote sensing data for increased comprehensiveness and accuracy. The results highlight the critical need for conservation initiatives and sustainable land-use policies in order to lessen the growing risks to mangrove ecosystems and their priceless roles in maintaining biodiversity around the world and regulating climate change.

1. Introduction

Mangrove forests are a type of unique coastal environment that thrive in the warm, wet conditions of the tropics and subtropics. These dense forests are home to salt-tolerant trees and shrubs that are well-adapted to life in the shifting conditions of coastal wetland ecosystems. (Mukhopadhyay et al., 2015). Mangrove forests provide numerous ecosystem services, including carbon sequestration, erosion control, and protection from storms and other natural disasters. Mangrove forests are important because they protect coastal soils from erosion and reduce the

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impact of waves and currents, as well as being home to a diverse diversity of plant and animal species. (Islam et al., 2019). Many people who live along the shore depend on the resources provided by mangrove forests and their way of life. These forests are home to many species of fish and shellfish with significant commercial value, which in turn sustain local fishing industries and guarantee food supplies for millions of people across the world (Rahman et al., 2013). Mangrove forests are important to the local economy because they provide lumber, medicinal plants, and other resources. Mangrove forests are important ecosystems that have ecological and social significance, but they are seriously threatened by human activities like pollution, changing land uses, and climate change. Due to the growing need for resources worldwide, millions of acres are being cleared for agricultural practices such as rice farming and shrimp aquaculture, which is one of the main drivers of mangrove deforestation (Aziz and Paul, 2015; Islam and Bhuiyan, 2018). The widespread devastation has far-reaching effects and exacerbates global warming since trees are essential for absorbing carbon dioxide. A large amount of greenhouse gas emissions is caused by deforestation, which releases stored carbon back into the atmosphere (Kundu et al., 2022). Deforestation has an impact on soil and water quality in addition to climate change. The loss of trees can result in problems including soil erosion, nutrient depletion, and decreased agricultural output because trees are essential for managing water cycles and avoiding erosion (Khan et al., 2021). Deforestation also tampers with local weather patterns, raising surface temperatures and decreasing evapotranspiration. Many plant and animal species are also threatened by the loss of forests, which could result in their extinction. This has a negative impact on the livelihoods of those that depend on forests in addition to endangering the world's food supply and water quality (Islam, 2014). Due to natural disasters including cyclones, tsunamis, and sea level rise, the Sundarbans mangrove forest in Bangladesh, which is recognized as a UNESCO World Heritage site, is especially susceptible to deforestation (Ali et al., 2021). Due to clear-cutting, encroachment, hydrological changes, chemical accidents, and climate change, the forest—which is essential for safeguarding coastal areas and supporting millions of people—is fast disappearing(Rahman et al., 2010). Non-timber forest products provide a substantial source of employment and income for almost 500,000 people living in the Sundarbans(Mahadevia and Vikas, 2019). The Sundarbans, which cover an area of around 10,000 km2, are the world's biggest mangrove forest, underscoring the need of its preservation for the sake of the environment and human welfare.

Remote sensing is essential for tracking and analyzing the dynamics of the mangrove forest cover. Although field inventories are crucial, remote sensing offers more effective information with benefits like cost-effectiveness, synoptic coverage, historical data availability, and recurrent observations (Islam et al., 2019; Kumar Sarkar et al., 2021). The utility of remotely sensed data is increased by recent advancements in satellite data processing hardware and software. By combining several variables for the detection of forest changes and evaluating their environmental impacts—such as precipitation, soil moisture, land surface temperature, and gas concentrations (CO2 and NO2) obtained from remote sensing techniques—this study seeks to close a research gap. The study makes use of Google Earth Engine (GEE), a cloud computing platform that provides remote sensing dataset data analysis toolkits. GEE's satellite image archives and processing tools make it easier to examine changes in the environment, including forest cover (Gorelick et al., 2017; Zhao et al., 2021). The CA-Markov model is used to estimate future changes and identify patterns based on the analysis of environmental data and forest change parameters (NDVI, EVI, MSAVI, and LULC). To sum up, this research tackles the crucial problem of mangrove destruction, concentrating on the Sundarbans, an essential ecosystem that is facing immediate danger. Through the application of Google Earth Engine capabilities and remote sensing data, the research endeavors to evaluate forest dynamics and develop associations with environmental aspects. In order to preserve mangrove forests and its priceless benefits to the environment and human well-being, it is imperative that conservation efforts be guided by the findings, which also serve to promote sustainable land use practices. The principal objectives this study aims to achieve are:

1. To identify the change and loss of forest area over the years and its impact on the environment
2. To forecast and model vegetation indices and environmental factors to view future trends
3. To assess the correlation among forest change over the years with the derogatory impacts

2. Methods and materials

2.1. Description of the study area

The largest continuous mangrove forest in the world, the Sundarbans is situated in the southwest of Bangladesh between the rivers Harinbanga and Baleshwar, which finally flow into the Bay of Bengal (Cornforth et al., 2013). A brief preview of The Sundarbans is presented in Table 1 along with a map of The Sundarbans in Fig. 1.

2.2. Description of data

Land imaging from moderate-resolution Earth-observing satellites, such as Landsat, provides an essential and irreplaceable tool for observing land use and land use change at these scales. Landsat's space-based land imaging is critical because it gives researchers and managers with repetitive and synoptic images of the Earth that would otherwise be unavailable to them (Kumar et al., 2020). Human health, agriculture, climate, energy, fire, natural catastrophes, urban growth, water management, ecosystems and biodiversity, and forest management all benefit from Landsat data (Rahman et al., 2013). Landsat 5 and 8 datasets with a spatial resolution of 30 m (Thakur et al., 2021) were used in this study for forest change monitoring and Salinity Index. The Landsat images were gathered between 2004 and 2022 and were in the WGS 1984 Geographic Reference Systems. They were used to measure temporal changes in mangrove cover as well as identify salinity variations in the soil. Landsat 5 was used up until 2012 and Landsat 8 was used here after. Landsat 7 was not used in the study due to complications regarding the scanline error often present in Landsat 7 scenes. Most images collected and used were having cloud cover less than 1 % with some Landsat 5 images having less than 5 % cloud coverage. Forest Change parameters were calculated monthly and annual Time Series

Table 1
A brief profile of The Sundarbans.

Attribute	Details
Location	Southwest Bangladesh between the rivers Harinbanga and Baleshwar, flowing into the Bay of Bengal
Size	10,000 km2
Coordinates	Latitudes 21° 27′ 30″ to 22° 30′ 00″ North, Longitudes 89° 02′ 00″ to 90° 00′ 00″ East
Distribution	60 % in Bangladesh, 40 % in India
Geographical Features	Tidal rivers, mudflats, mangrove forests
Ecological Phenomena	Monsoonal rains, flooding, delta building, tidal influence, mangrove colonization
Climate	Rainfall up to 2800 mm, monsoon season from June to October, storms, cyclones, tidal surges
Economic Activities	Lumber and fuel wood extraction, fishing, honey and other forest product collection
Conservation Designations	Three animal sanctuaries, one national park, all designated World Heritage Sites by UNESCO
Population Dependency	Over 2.5 million people in villages bordering the Sundarbans, heavily reliant on forest products

(Chatterji, 1994; Hussain and Acharya, 1994; Mukul et al., 2020; Uddin et al., 2013).

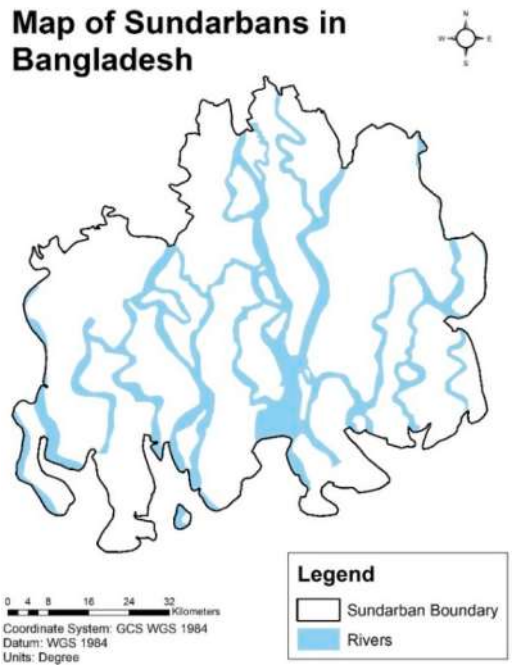


Fig. 1. Study Area.

chart was generated.

Moderate Resolution Imaging Spectroradiometer (MODIS) dataset with a spatial resolution of 1 km and acquisition date from 2004 to 2022 was used to derive Aerosol Optical Depth and Land Surface Temperature (Justice et al., 2002). MODIS (or Moderate Resolution Imaging Spectroradiometer) is a critical instrument aboard the Terra (formerly EOS AM-1) and Aqua (formerly EOS PM-1) spacecraft (Salomonson et al., 1989). Terra's orbit around the Earth is timed so that it crosses the equator from north to south in the morning, while Aqua crosses the equator from south to north in the afternoon. Terra MODIS and Aqua MODIS scan the whole Earth's surface every 1 to 2 days, collecting data in 36 spectral bands (see MODIS Technical Specifications) (Zhang et al., 2003). This data will help us better comprehend global dynamics and processes that occur on land, in the oceans, and in the lower atmosphere. MODIS is playing an important role in the creation of validated, global, interactive Earth system models capable of reliably predicting global change sufficiently to assist policymakers in making smart decisions about environmental protection.

GEOS or Geostationary Operational Environmental Satellite Network provides Carbon Monoxide concentration, Nitrogen Dioxide concentration, Climate Water Deficit and Soil Heat Flux data obtained through Modern-Era Retrospective analysis for Research and Applications version 2 (MERRA-2) dataset (Buchard et al., 1980). The Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2) provides data beginning in 1980 and was introduced to replace the original MERRA dataset because of the advances made in the assimilation system that enable assimilation of modern hyperspectral radiance and microwave observations, along with GPS-Radio Occultation datasets (Gueymard and Yang, 2020). MERRA-2 is the first long-term global reanalysis to assimilate space-based observations of aerosols and represent their interactions with other physical processes in the climate system.

FLDAS: Famine Early Warning Systems Network (FEWS NET) Land Data Assimilation System dataset includes a number of land surface parameters that were simulated using the Noah 3.6.1 model (Loeser et al., 2020).The data span from January 1982 to the present and have a 0.10 degree resolution. The worldwide (60S, 180 W, 90 N, 180E) spatial coverage and monthly temporal resolution are both used but the worldwide monthly dataset has taken the place of the FLDAS regional monthly datasets, which are no longer accessible (Pervez et al., 2017).

Precipitation data obtained using Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) dataset which is also under GEOS or Geostationary Operational Environmental Satellite Network satellite of NASA which is a 35 + year quasi-global rainfall data set. Spanning 50°S-50°N (and all longitudes) and ranging from 1981 to near-present (Dinku et al., 2018). CHIRPS incorporates NASA's in-house climatology, CHPclim, 0.05° or 11 km spatial resolution satellite imagery, and in-situ station data to create gridded rainfall time series for trend analysis and seasonal drought monitoring (Katsanos et al., 2016).

Table 2 contains a short overview of every data used along with relevant information. All parameters were collected annually, and annual Time Series chart was also generated. Masking of clouds and cloud shadows was based on the "pixel quality attributes" band using CloudMask algorithm in Google Earth Engine. Both MODIS and Landsat datasets were cloud masked before performing any analysis.

2.3. Analysis methods

Most analysis of the study heavily relied on Google Earth Engine platform which incorporates a planetary-scale analysis framework with a multi-petabyte collection of satellite pictures and geographical datasets. Earth Engine is used by scientists, researchers, and developers to track trends, quantify differences, and identify areas of change on Earth's surface (Gorelick et al., 2017). Forest change detection, deriving environmental parameters, creation of time series all were done through Google Earth Engine. R-Studio was used in forecasting of the parameters which was based on Autoregressive Integrated Moving Average model which essentially creates a linear equation which describes and forecasts time series data (Kalpakis et al., 2001). It predicts future values based on past values using lagged moving averages to smooth time series data and predicts by looking at the deviations rather than the actual values inside the series and when making predictions, RMSE or Root Mean Square Error and MASE or Mean Absolute Scaled Error was also calculated using R-studio. Identification of trend and type of trend was also performed using R-studio. It was at the very end used to find out the correlation coefficients and lagged correlation among forest change and environmental indicators. IDRISI Selva which offers almost 300 modules for the analysis and presentation of digital spatial information and serves as an integrated GIS and Image Processing software solution (Hamdy et al., 2017) was used to effectively model the forest change indicators using the CA-Markov Model with influential parameters being air quality parameters, DEM, LST and Precipitation as these cause the highest level of interference with vegetation health. Jupiter Python was then used to calculate the Receiver Operating Characteristics (ROC) and Area Under Curve (AUC) for validation.

2.3.1. Land-use land-cover (LULC) change detection

The systematic examination and classification of human and natural landscape effects across time using scientific and statistical methodologies is called "Land Use and Land Cover" (LULC) (Attri et al., 2015). Land usage and land cover are different yet often used interchangeably. Land cover includes vegetation, buildings, water, soil, and other terrain and informs thematic mapping and change detection analyses (Singh et al., 2015). "Land use" describes how a property is used, such as a park, farm, or wildlife refuge.

LULC change detection was done entirely using Google Earth Engine Platform through supervised classification using the smileCart classifier annually from 2004 to 2022. In total 800 training samples were taken and categorized into four LULC classes: Water Body, Forest Area, Bare Soil and Built-up Area. Time series of the LULC classes was derived using ui.Chart.image.series function and the areas of each class was calculated.

Table 2
Description of data used.

Derived Parameters	Satellite	Sensor	Acquisition Date	Spatial Resolution	Temporal Resolution
Vegetation Loss Rate, Normalized Difference Salinity Index (NDSI)	Landsat 5, 7 & 8	OLI, TIRS, TM, ETM+	2004–2022	30 *m*	16 days
Aerosol Optical Depth (AOD)	Aqua and Suomi NPP JPSS	Moderate Resolution Imaging Spectroradiometer (MODIS)	2004–2022	1 km	Daily
Land Surface Temperature	Aqua	Moderate Resolution Imaging Spectroradiometer (MODIS)	2004–2022	1 km	Daily
CO, NO2, CWD, SHF	GEOS	MERRA-2	2004–2022	11 km	Daily
Soil Moisture	GEOS	FLDAS: Famine Early Warning Systems Network (FEWS NET) Land Data Assimilation System	2004–2022	11 km	Monthly
Precipitation	USGS EROS	CHIRPS	2004–2022	11 km	Monthly

2.3.2. *Modified Soil Adjusted Vegetation Index (MSAVI), Normalized Difference Vegetation Index (NDVI) and Enhanced Vegetation Index (EVI) change detection*

The Modified Soil-Adjusted Vegetation Index (MSAVI) is an alternative to the normalized difference vegetation index (NDVI) and normalized difference chlorophyll index (NDCI) in cases where there is either insufficient vegetation or insufficient chlorophyll in the plants (Qi et al., 1994). MSAVI was derived monthly using the already established expression of MSAVI.

$$MSAVI = \frac{2\mathrm{NIR} + 1 - \sqrt{(2NIR + 1)^2 - 8(\mathrm{NIR} - \mathrm{Red})}}{2}$$

*NIR = Near Infra-Red (Wang et al., 2006)

When analyzing remote sensing readings, usually from a space platform, the Normalized Difference Vegetation Index (NDVI) provides a quick visual assessment of whether or not the observed target has living, green vegetation (Pettorelli et al., 2005). NDVI also has its own established expression and was derived monthly.

$$NDVI = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}}$$

*NIR = Near Infra-Red (Wang et al., 2006)

The Enhanced Vegetation Index (EVI) is an 'optimized' vegetation index that aims to improve vegetation monitoring by de-coupling the canopy background signal and reducing the effects of the atmosphere on the vegetation signal in high biomass zones (Jarchow et al., 2018). The established expression of EVI is,

$$EVI = 2.5^* \frac{(\mathrm{NIR} - \mathrm{Red})}{(\mathrm{NIR} + \mathrm{C1}^*\mathrm{Red} - \mathrm{C2}^*\mathrm{Blue} + \mathrm{L})}$$

*NIR = Near Infra-Red, C1 = 6, C2 = 7.5, L = 1 (Li et al., 2018)

The indexes were all derived using GEE and annual Time series was generated using ui.Chart.image.series function.

2.3.3. *Environmental impact factors and time series*

Most parameters were derived through GEE using available and ready datasets. Some parameters were obtained from the official website of NASA Giovanni (Giovanni - Data Selection (nasa.gov)) which is a NASA Goddard Earth Science Data and Information Services Center (GES DISC) Distributed Active Archive Center (DISC) web application that provides a simple, intuitive way to visualize, analyze, and access Earth science remote sensing data, particularly from satellites, without having to download the data . Normalized Difference Salinity Index (NDSI) was calculated using the following expression.

$$NDSI = \frac{(\mathrm{Green} - \mathrm{SWIR})}{(\mathrm{Green} + \mathrm{SWIR})}$$

*SWIR = Short Wave Infra-Red (Asfaw et al., 2018)

Annual Time Series was derived using similar function as the forest change detection parameters.

2.3.4. *Mann-Kendall trend Test and Sen's Slope*

The Mann-Kendall Trend Test (also known as the M−K test) examines time series data for a monotonic increase or decrease in Y values (Mondal et al., 2012). This test was used to determine the presence of trend in the time series data derived where p-value of less than 0.05 depicts a presence of trend.

When looking for patterns in univariate time series, Sen's slope estimator can be helpful and outliers tend to have little effect on it (Gocic and Trajkovic, 2013). Whether the value of slope is negative or positive represents if the trend present is decreasing or increasing respectively. These statistical analyses were both performed for forest change detection parameters in R-studio after importing the time series data as CSV.

2.3.5. *Annual co-efficient of variation*

). Dispersion between data sets of different types can be compared using the measure (Brown, 1998). This was performed to see the variation of forest cover over the years based on the four forest change parameters using R-Studio.

2.3.6. *CA Markov model*

A useful technique in land-use change modelling is the CA-Markov model, which combines a Markov chain and a spatially specified cellular automaton. By considering the transition probabilities between various land-use categories, it makes it possible to simulate and anticipate how land-use will vary over time (Subedi et al., 2013). Using its capacity to accurately replicate temporal and spatial patterns, the CA–Markov model combines Markov theories with cellular automata (CA) to forecast changes in land use over time and space. Utilizing conversion tables and conditional probability from the conversion map, the CA–Markov module in IDRISI32 combines Markov processes and cellular automaton filtering to anticipate stages of land use change. The CA-Markov model, which is incorporated into the IDRISI program, simulates changes in land use by adhering to a particular procedure. Using Markov chain analysis, transition rules are first established by projecting future modifications based on observed transition probabilities across land use maps in various years. The transfer area shows the anticipated changes in land use in the upcoming period, while the transition probability matrix shows the likelihood of different land use types changing to others. The 2004 map is superimposed on top of the 2022 land use map as the baseline. The transformation rules for CA-Markov simulations are derived from this matrix. In order to generate a space weighting factor that is influenced by the surrounding cellular state, CA filters are next determined. In this investigation, a typical 5×5 contiguity filter is employed. In order to start the simulation process, the beginning point and the number of CA iterations are finally decided. This all-encompassing method makes it easier to analyze land use changes in depth while taking temporal and spatial factors into account.

2.3.7. Forecasting of time series and modelling of forest change parameters

Auto Regressive Integrated Moving Average model was fundamentally utilized using R-Studio to forecast/predict the time series data of forest change parameters as well as environmental impact indicators for the next 8 years until 2030. The spatial patterns of LULC and Vegetation indexes were simulated using the CA-Markov Model which is useful for stimulating spatial–temporal patterns because it integrates the best features of both the Markov and CA theories, namely their time series and spatial predictions (Singh et al., 2015). The CA-Markov model considers not only the desirability of land use or vegetation index changes, but also the impact of environmental, social, and economic aspects. The simulation was done using Idrisi Selva and influential parameters effecting spatial patterns were air quality, DEM, LST and precipitation.

2.3.8. Auto Correlation/Lagged correlation

The term "autocorrelation" is used to describe the degree of similarity or correlation between the values of the same variable or data series over time intervals (Moews et al., 2019). This measure of correlation is sometimes referred to as a lagged or serial correlation coefficient. Lagged Correlation was performed to identify the relations between Forest change dynamics and environmental impact parameters and whether the relation is forward or backward i.e. If forest change occurs first or environmental impacts causes forest change and if past changes have any effect on future impacts. The value ranged from −1 to + 1 where if the value lies from −1 to 0 then it is a negative correlation and values between 0 to + 1 indicating positive correlation. The correlation coefficients were determined using Cor function in R-studio and lagged correlation was also performed between the time series data.

2.3.9. Accuracy assessment and validation

For accuracy assessment of LULC classification,random test points were manually taken and Producer's Accuracy, User's Accuracy, Overall Accuracy and Kappa coefficient was calculated using ArcGIS and Google Earth. Root Mean Square Error (RMSE) and Mean Absolute Scaled Error (MASE) was calculated using R-studio to find the accuracy of forecast models. A value of MASE lower than 1 Mean, 2024 and for RMSE a value lower than 0.5 (RMSE, 2024) is universally regarded as an accurate model. The validation metrics of Receiver Operating Characteristics (ROC) and Area Under the Curve (AUC) were then computed using Jupiter Python.

2.3.10. Methodological framework

.A quick recap of the entire workflow is visualized in the methodological flow chart as illustrated in Fig. 2.

3. Results

3.1. Temporal variation of land use cover for forest change detection

3.1.1. Land use land cover change detection

Fig. 3 depicts the change in LULC pattern from 2004 to 2022 for Sundarbans divided into 4 principal classes and Table 3 denotes the corresponding area of classes over the years.

During the period beginning in 2004 and concluding in 2022, we analyzed water bodies, forest area, bare soil, and built-up area. The research findings shed light on the temporal variations and trends observed in these characteristics. Throughout the examined time period, the average water coverage was approximately 2450 square kilometers.

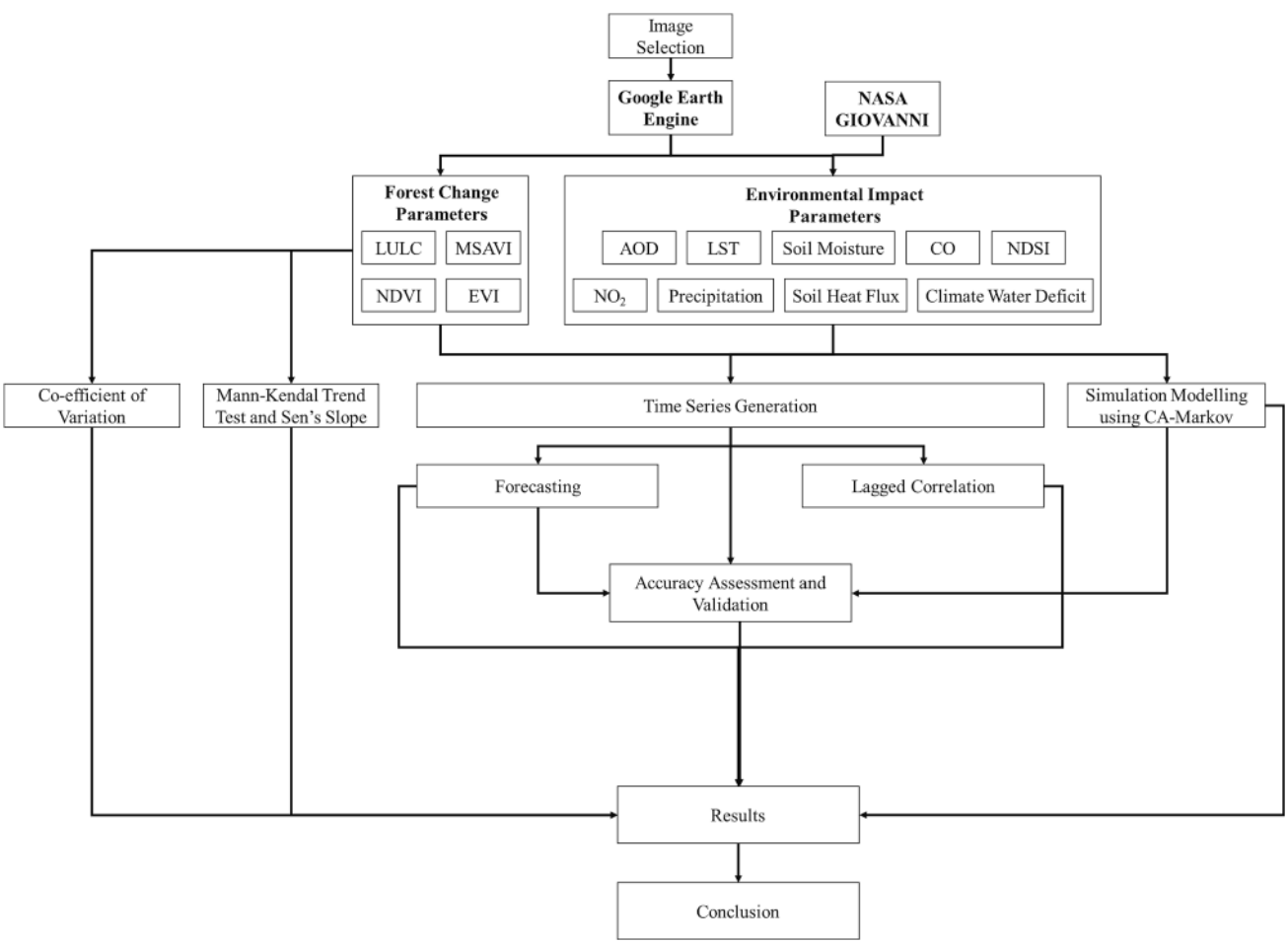


Fig. 2. Methodological Framework.

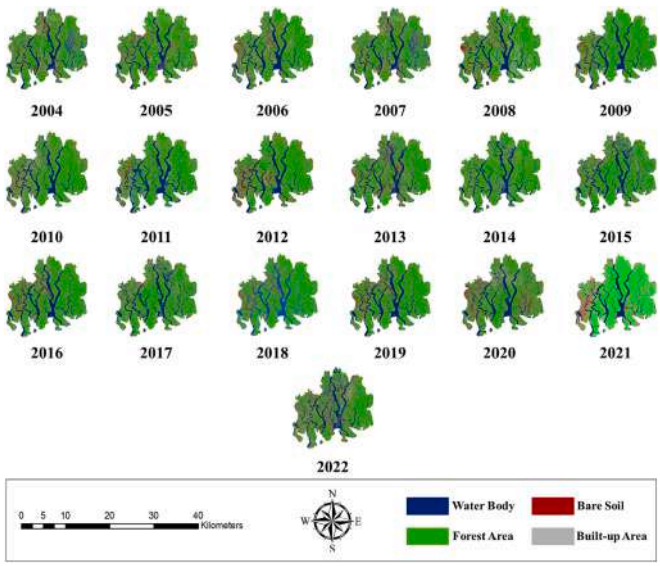


Fig. 3. Spatial variation of LULC classes.

Table 3
Area of LULC Classes by year.

Year	Water Body	Forest Area	Bare Soil	Built-up Area
2004	2500.60	3531.16	368.77	95.11
2005	2446.1	3396.6	556.57	96.37
2006	2495.75	3471.82	409.48	118.60
2007	2495.02	3098.13	781.98	120.52
2008	2495.62	3304.67	574.88	120.48
2009	2421.96	3395.20	576.51	101.99
2010	2411.44	3365.58	590.13	128.49
2011	2328.44	3468.97	577.05	121.19
2012	2334.72	3439.25	529.55	192.13
2013	2496.24	3204.10	618.77	176.54
2014	2411.00	3272.28	661.16	151.21
2015	2454.03	3361.73	508.32	171.57
2016	2492.80	3265.15	574.90	162.79
2017	2502.92	3287.44	575.65	129.64
2018	2246.62	3353.87	673.35	221.80
2019	2449.17	3139.59	686.69	220.19
2020	2477.82	3187.60	611.90	218.33
2021	2445.97	3198.38	618.66	232.64
2022	2488.79	3065.80	710.87	230.17

This indicates that the distribution of water bodies in the region has remained relatively stable, with sporadic fluctuations. Nevertheless, there may have been regional variations or isolated incidents during this time period.

The study shows that the area's forest cover has significantly decreased over the studied period. From its maximum of 3531.16 square kilometers in 2004 to 3065.80 square kilometers in 2022, the forest area has decreased at a 2.66 % yearly rate. It is most likely due to a variety of forest-related risks, such as deforestation, changes in land use, and natural catastrophes such as Cyclone Sidr in 2007; Aila in 2009; and a forest fire in 2019. Simultaneously, there is a noticeable upward trend in measurements of bare soil, which have increased from 368.77 to 710.87 square kilometers after 18 years. Natural erosion, land management techniques, mining and building activities, and agricultural methods are a few examples of the variables that could cause these alterations. To fully understand the underlying causes and implications, additional study as well as a thorough analysis of local land use and management techniques are required. The analysis also shows that during the course of the study, the amount of developed land in the area increased. A combination of factors including population growth, the creation of new sawmills and fisheries, and infrastructural improvements have caused the built-up area to increase from 95.11 sq. km in 2004 to 230.17 sq. km by 2022. Concerns have been raised concerning this expansion's effects on natural ecosystems, the disappearance of open spaces, increased pollution, and possible conflicts with other land uses. Sustainable development strategies, including wise growth plans, efficient land-use planning, and the protection of green areas, are crucial for resolving these issues and achieving a balance between environmental preservation and human needs.

3.1.2. *Forest area Forecasting from LULC and simulation*

Forecasting of forest area is imperative to know the future condition of the forest based on present and past conditions. This will help us be better prepared to face what is ahead and take suitable decisions accordingly to lessen the impacts and extent of damage. Fig. 4 shows the forecasted results up until the year 2030.

ARIMA (Auto Regressive Integrated Moving Average) model was used to make forest-specific forecasts as we are only keen on delineating deforestation, that factor in both historical patterns and current values. There is a discernible downward trend in the forest cover from 3531.16 sq. km in 2004 to 2846.97 sq. km in 2030, as shown in the projected chart. Over a span of 26 years, the forecast indicates a significant decrease of 684.19 sq.km in the forest area. This represents approximately a 19 % decline in the total forest area. The forecasted values suggest that if the current trends continue, the forest area will experience a notable reduction over the specified time period. This decline calls into question the long-term health of the forest ecosystem. Loss of habitat, dwindling biodiversity, and rising carbon emissions are just some of the biological and environmental effects that can result from a drop-in forest area (Naikoo et al., 2023). The predicted decline in forest cover highlights the critical importance of preventative measures to preserve and restore forest ecosystems. Fig. 5 depicts a simulated model of the Sundarbans LULC in 2030. Area for each category is displayed in Table 4, which is provided alongside the model.

Significant changes in a number of land cover classes are revealed by simulation modeling. The considerable increase in urban areas and the corresponding steady decrease in forested areas are particularly noticeable. Natural ecosystems and green spaces are being lost as a result of the increase in built-up regions brought about by urbanization, infrastructural expansion, and population growth. (Talukdar et al., 2020). Furthermore, a rise in bare soil coverage indicates land clearing, agricultural practices, or vegetation.

3.2. *Temporal variation of MSAVI for forest change detection*

3.2.1. *Spatial and annual variation of MSAVI*

Fig. 6 shows monthly spatial variation of MSAVI in the form of index anatomy and Fig. 7 depicts the Annual time series of MSAVI fluctuations.

The time-series study of the Modified Soil-Adjusted Vegetation Index (MSAVI) provides useful information on the index's fluctuation. MSAVI averages out to 0.19, with a low of 0.12 and a high of 0.19. For the most part, the index is able to capture a diversity of plant health states within this range.

By looking at seasonal trends, you can see that MSAVI values are often highest in March and lowest in November. Seasonal shifts have an effect on plant growth and vigor, which is shown in this cyclical pattern. Changes in vegetation dynamics may help explain the apparent increase in MSAVI value variability observed between 2013 and 2017. Significant occurrences, such Cyclone Sidr in 2007, are associated with a dramatic decline in MSAVI scores. Natural disasters can cause severe damage to vegetation, which in turn leads to poorer vegetation health and lower MSAVI values, therefore this reduction is to be expected. Moreover, intriguing tendencies emerge when analyzing MSAVI data over time. The year 2005 has the greatest MSAVI value, suggesting that vegetation health was at an all-time high. On the flip side, the lowest MSAVI value was reported in 2007, which is consistent with the

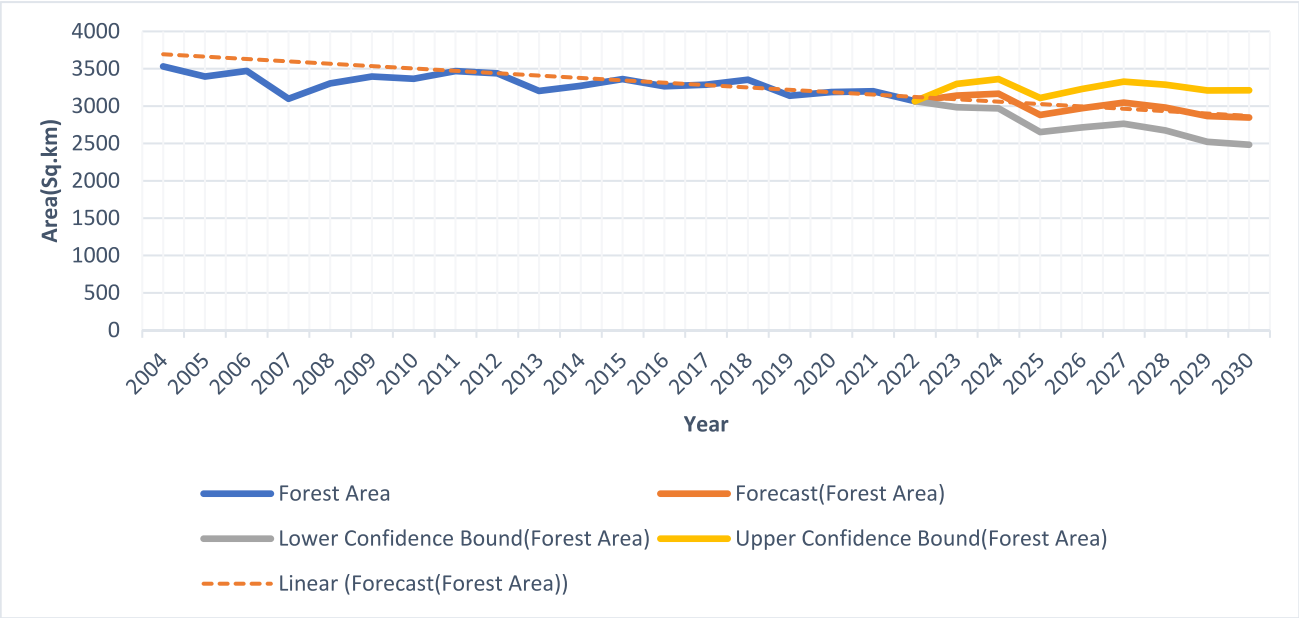


Fig. 4. Forecasting of Forest Area using ARIMA. * RMSE: 0.48; MASE: 0.51

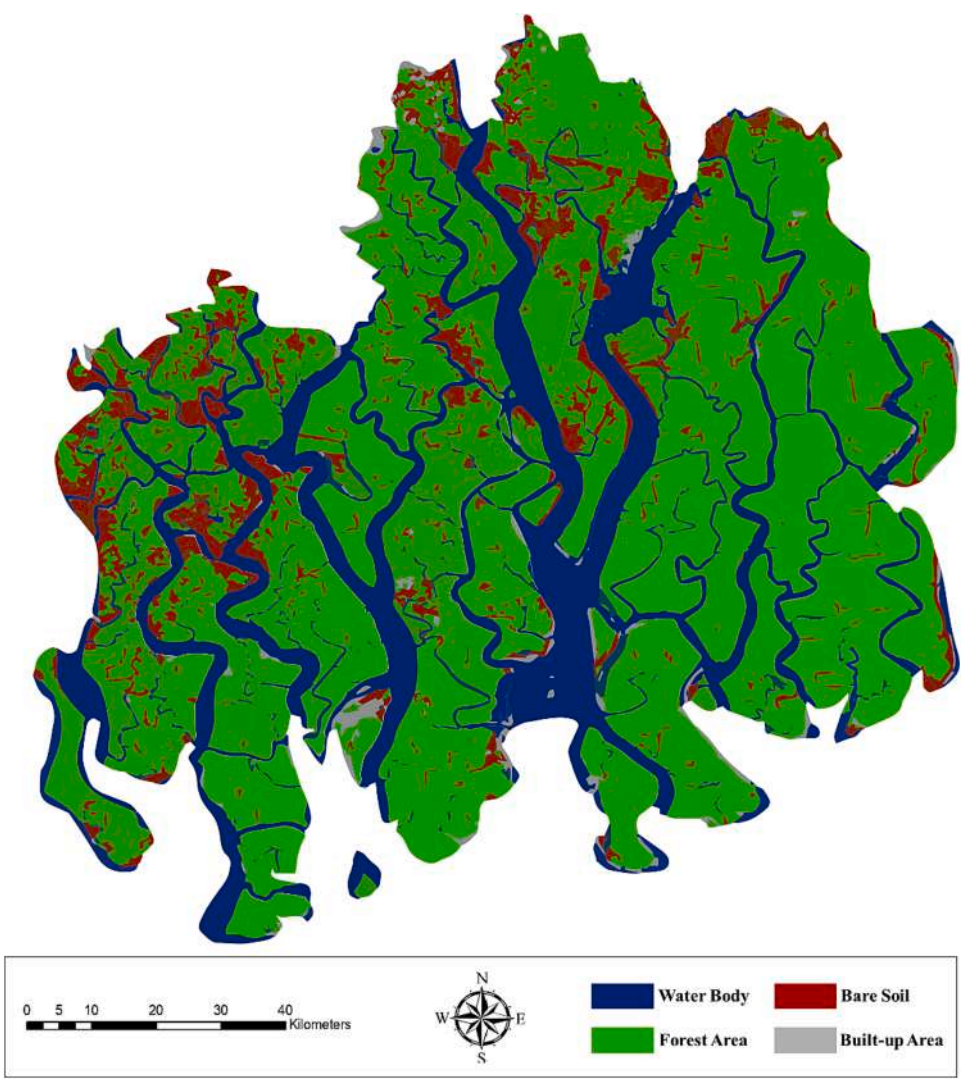


Fig. 5. Spatial Simulation of LULC for 2030.

Table 4
Area of LULC class based on simulation.

Land Use	Area (Sq.km)
Water Body	2495.50
Forest Area	2946.97
Bare Soil	763.74
Built-up Area	294.43

influence of cyclone Sidr.

It is clear that the MSAVI levels have dropped significantly by 2022 when compared to the base year of 2004. This decrease may be indicative of declining plant health over the time frame under consideration. Additional research is needed to determine the causes of this loss in plant life, which may include changes in land use, climatic conditions, or human activities that have an effect on plant cover. As evident from Table 5 which shows the results of Mann-Kendall Trend test and Sen's Slope MSAVI values shows a presence of negative trend and so further forecasting can help in estimating the anticipated value of MSAVI i.e., forest cover health in the future.

3.2.2. Forest area Forecasting from MSAVI and simulation

Much like forest area of LULC, MSAVI was forecasted using ARIMA as shown in Fig. 8. The forecast shows a decreasing trend in MSAVI values indicating towards lower vegetation in future years. The Spatial simulation using CA-Markov in Fig. 9 shows the spatial distribution of MSAVI in 2030.

The forecast data shows the Modified Soil-Adjusted Vegetation Index (MSAVI) values for 2023–30. Vegetation health declines as MSAVI values fall over the projected period. 2023′s MSAVI is 0.140723734 which indicates modest plant health. MSAVI readings have steadily declined since then. The number decreases to 0.139610295 in 2024, 0.138496857 in 2025, and 0.137383418 in 2026. These decreases suggest a gradual decline in vegetative health. Later years of the forecast show a further drop with MSAVI declining to 0.126269979 in 2027, indicating a substantial vegetative health decline and 0.125156541, 0.124043102, and 0.122929663 follow in 2028, 2029, and 2030, respectively. The prognosis shows that MSAVI vegetation health is expected to drop progressively throughout the anticipated years. Changes in land use, climate, or human activity could cause this drop. To apply proper conservation and management strategies, this decline in vegetation health must be monitored and assessed. The deteriorating condition of vegetation health is also evident as seen in the spatial distribution showing more yellow regions than green. Understanding these changes can improve land use planning, ecosystem restoration, and sustainable resource management to reduce degradation and promote vegetation health and resilience in the studied region.

3.3. Temporal variation of NDVI for forest change detection

3.3.1. Spatial and annual variation of NDVI

Fig. 10 shows monthly spatial variation of NDVI in the form of index anatomy and Fig. 11 depicts the Annual time series of NDVI fluctuations.

The average NDVI value in this scenario varies from 0.60 on the low end to 0.76 on the high end. We can infer from these numbers that the density of vegetation is somewhere between average and high. NDVI shows clear seasonal variation, with the highest values appearing in April and the lowest appearing in October. Temperature, precipitation, and solar radiation are only a few of the variables that could affect this trend. In many parts of the world, April is the beginning of spring, a time of longer days and milder weather that is ideal for plant development and, consequently, higher NDVI readings. In contrast, October is typically the beginning of the autumn season or the beginning of the dry season, both of which result in lower NDVI values as vegetation begins to senesce or faces water stress. The variation between the NDVI values in 2011 and 2016 is larger than in any other time period studied. Interannual climate variability, changes in land cover, and human activities like deforestation and land use conversion all play a role in this shift. Note that NDVI can be severely impacted by extreme occurrences. The greatest NDVI value was observed in 2005, which could be indicative of remarkable vegetation growth brought on by conducive conditions like copious rainfall or an appropriate temperature. The year 2007 saw the lowest NDVI rating, which is due to the widespread devastation that Cyclone Sidr inflicted on the region's vegetation. Like the Modified Soil-Adjusted Vegetation Index (MSAVI) values, the annual mean NDVI has been going down. The NDVI trended downward from 0.80 in 2004 to 0.64 in 2022 which is evident in Table 6 showing results of Mann-Kendall Trend test and Sen's Slope. Climate change, soil degradation, urbanization, and altered land management practices are just a few of the possible causes of this loss in plant life.

3.3.2. Forest area Forecasting from NDVI and simulation

Fig 12 shows the forecasted values of NDVI using ARIMA model and spatial simulation of NDVI in 2030 is shown in Fig. 13.

The values for NDVI between the years 2023 and 2030 are shown in the provided graph. The graph displays a downward trend in NDVI readings over the time frame chosen. The NDVI begins to decline after 2023, when it is at 0.66, and continues to do so every year after that. After dropping to 0.65 in 2024, 0.64 in 2025, 0.63 in both 2026 and 2027, and finally 0.62 in 2028, the value continues to decline steadily thereafter. In 2029, the NDVI will be 0.61, and by 2030, it will have levelled off at 0.60. The declining NDVI values over time are indicative of diminishing plant health and density. Urbanization, farming, and other changes in how land is used could potentially have an effect. Natural ecosystems may be converted into urban areas or intensively managed agricultural regions, both of which may have less vegetation cover and biodiversity. Such alterations to the landscape may play a role in the predicted drop in NDVI levels. Understand that regional and local factors particular to the study area can affect the actual numbers and their interpretations, and that this forecast table only provides a simplified picture of NDVI trends.

This decreasing trend is also evident in the spatial variation simulated using CA-Markov which shows more yellowish tint than pure green indicating less vegetation density specially around areas closer to human habitation indicating human interference, deforestation, agriculture and infrastructure development may be the cause.

3.4. Temporal variation of EVI for forest change detection

3.4.1. Spatial and annual variation of EVI

EVI is the final vegetation index used to understand forest change dynamics in this study. Similar to previous indexes, an index anatomy was generated to measure spatial variations in Fig. 14 and the annual time series in Fig. 15 gives us an estimate of annual EVI fluctuations.

In general, EVI levels can be anywhere from 0.36 on the low end to 0.45 on the high end in average. The EVI is a satellite-based index of plant health that quantifies how green an area is. EVI has shown a declining tendency with natural disasters like past indices have.

Higher EVI levels can be seen in February, while lower values can be seen in November, indicating a seasonal pattern in EVI values. Changes in plant growth and activity throughout the year may explain this trend. Higher EVI values can be expected in the month of February because this is when plants are normally at their vegetative peak. Lower EVI readings can be seen in November because plants go dormant as they age and reach senescence. In addition, between 2007 and 2017, there was more variation in EVI. Climate change, shifts in land use, and ecological upheavals are all possible contributors to the observed variation. In 2005, the EVI reached its maximum value, marking a time of unprecedented plant vitality and density. On the other hand, 2008 saw the lowest EVI score, a sign of less robust plant life. These outlandish figures may have anything to do with conducive weather circumstances or deliberate land management. This Mann-Kendall trend test and Sen's Slope further

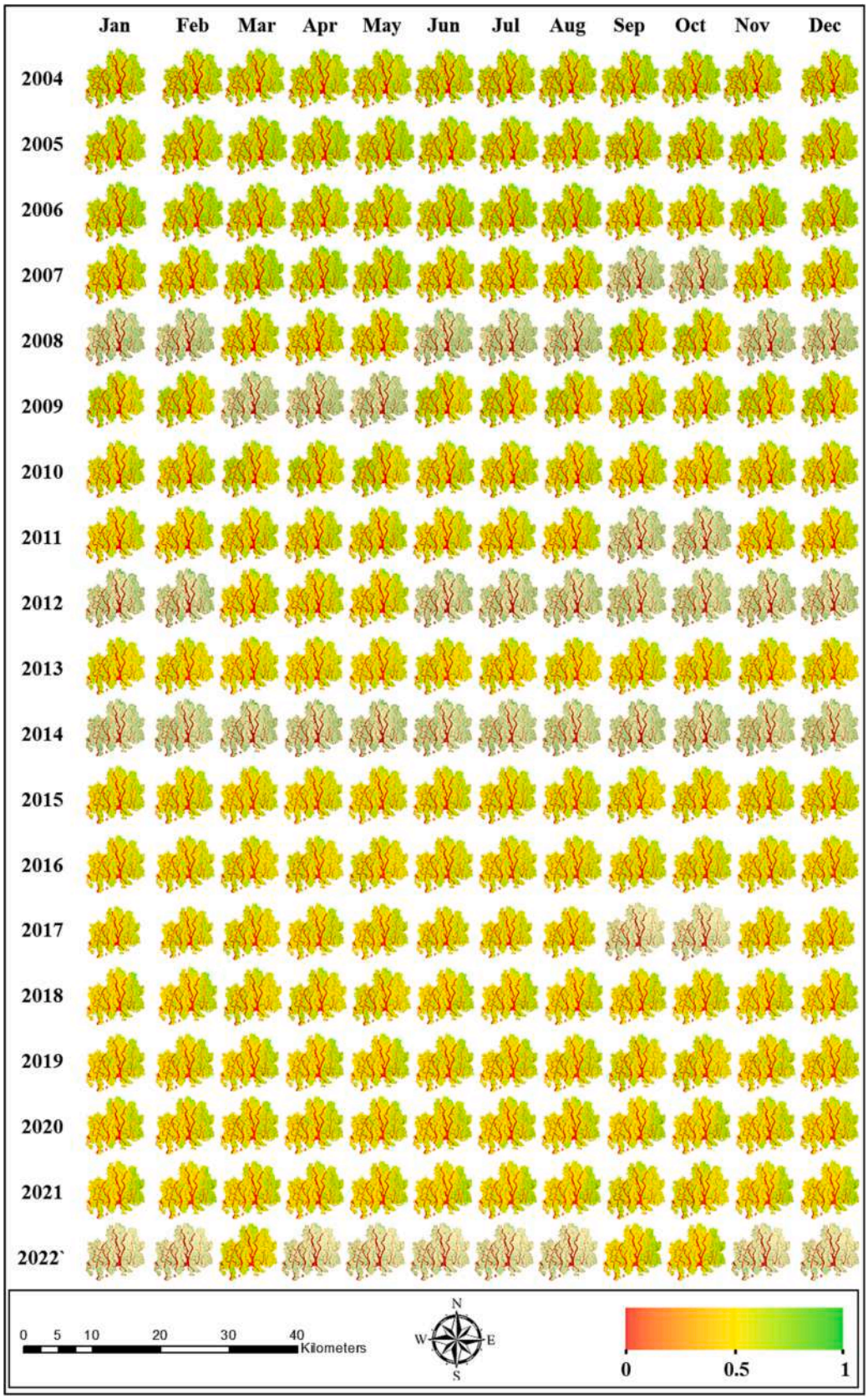


Fig. 6. Index Anatomy of Monthly Spatial Variation of MSAVI.

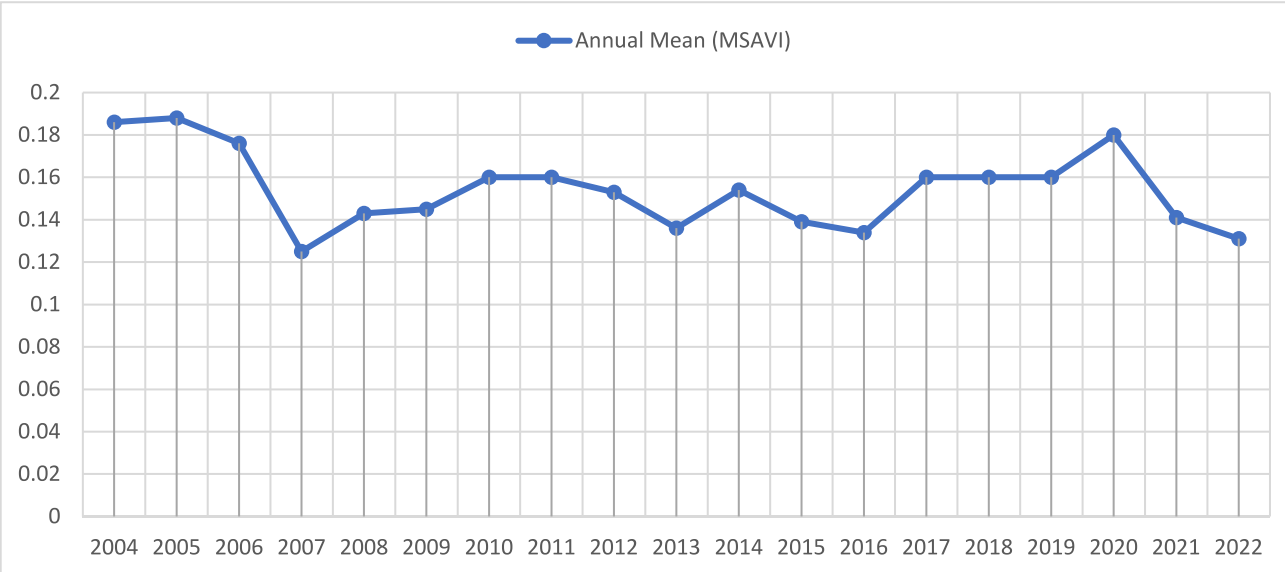


Fig. 7. Annual Time Series of MSAVI.

Table 5
Mann-Kendall Trend Test and Sen's Slope for MSAVI.

Mann-Kendall Test				
Index	Kendall's Tau	Variance	P-value	Presence of Trend
MSAVI	**0.421**	950	0.01037	Yes
Sen's Slopes				
Index	Sen's Slope (Q_i)	Sign of Q_i	Type of trend	Rate of growth
MSAVI	**−0.00425**	Negative	Decreasing	−0.00425/year

solidifies this decreasing trend as shown in Table 7.

3.4.2. *Forest area Forecasting from EVI and simulation*

Similar to past indexes ARIMA model was employed to obtain the forecast in Fig. 16 and CA-Markov model was used to simulate the spatial variation of EVI in 2030.

The Enhanced Vegetation Index (EVI) forecast chart shows future values from 2023 to 2030. The pattern of the EVI values decreasing over time indicates that the greenness and health of the vegetation are deteriorating. From a predicted 0.374 in 2023, the EVI is expected to drop to 0.370 in 2024, showing a gradual weakening of plant life. The following years see even lower EVI values, with 2025 seeing 0.365, 2026 seeing 0.361, 2027 seeing 0.356, 2028 seeing 0.352, and 2029 seeing 0.347. These diminishing numbers point to a slow but steady decline in the quality of plant life and canopy cover. The lowest estimated EVI is 0.343, which is expected to be reached in 2030. This is a further sign of a reduction in plant vitality compared to past years. This projection hinges on the assumption that climate, land use, and disturbances, all of which have an impact on EVI will maintain their current levels of influence going forward. The spatial variation also shows more yellow patches in the map compared to green patches indicating towards a decrease in EVI value and forest health as shown in Fig. 17. It is possible that climate

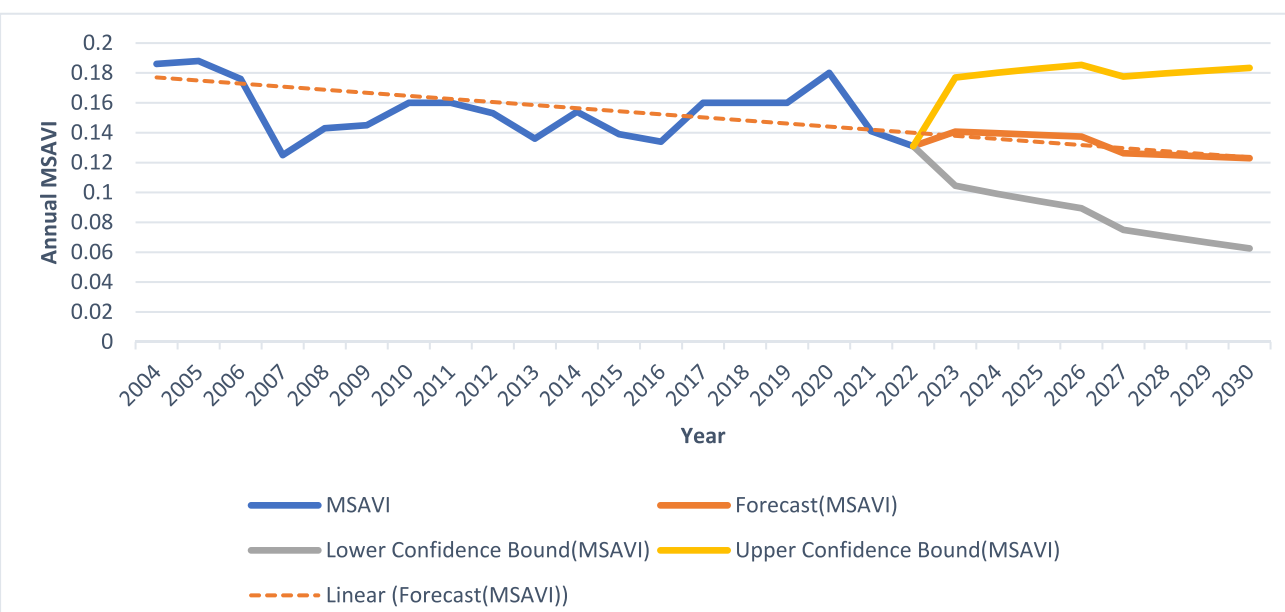


Fig. 8. Forecasting of MSAVI using ARIMA. *RMSE: 0.02; MASE: 1.02.

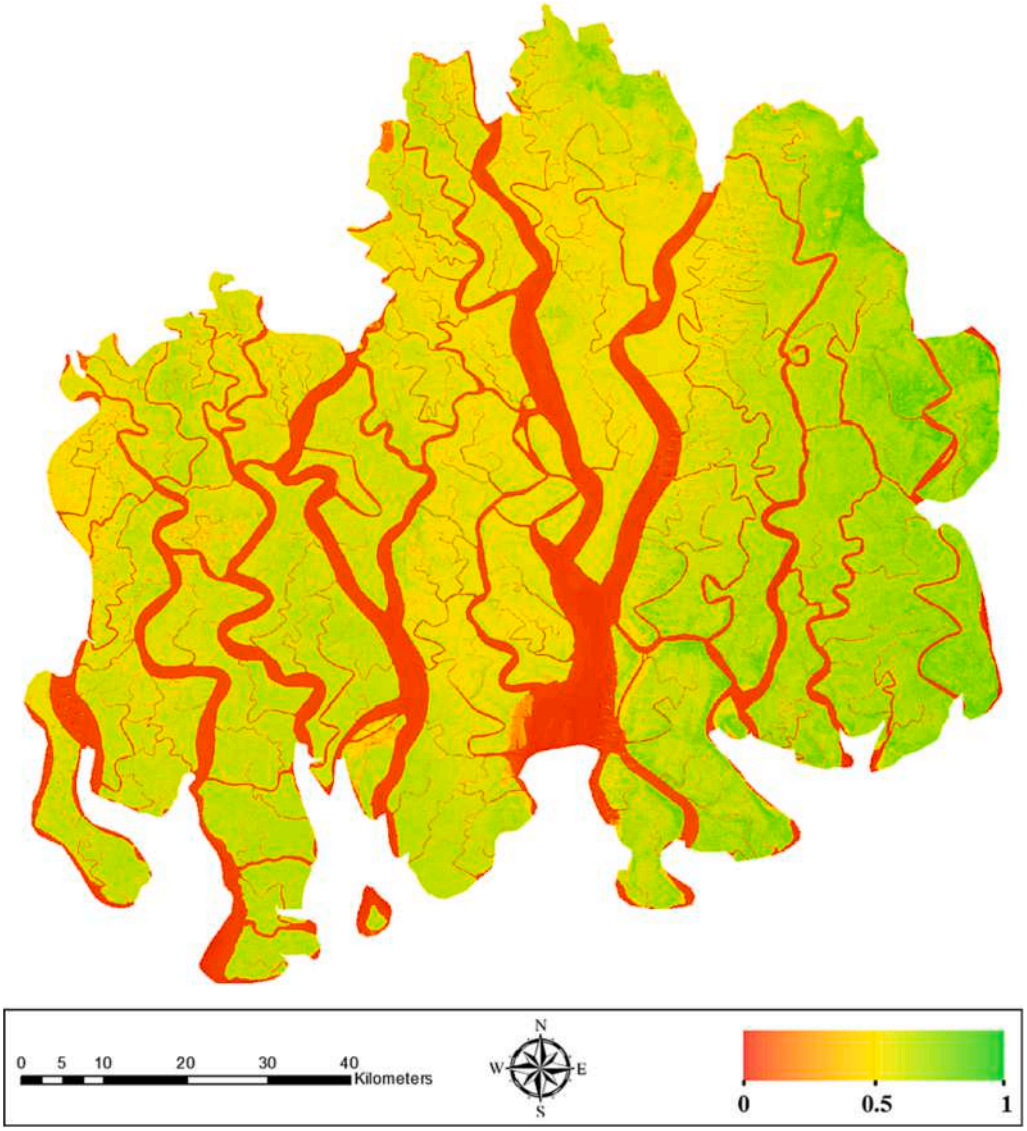


Fig. 9. Spatial Simulation of MSAVI for 2030.

change, land degradation, or changes in vegetation patterns are contributing to the predicted long-term decline in EVI values. These alterations may have an effect on water supply, nutritional availability, and other factors essential to plant growth, which could reduce EVI. Trends in EVI monitoring and analysis over time can reveal important information about plant health and dynamics. Ecosystem health, biodiversity, and land management practices may be affected if the predicted fall in EVI values is verified. Understanding the precise variables contributing to the observed and predicted trend will require additional research into the origins of this reduction.

3.5. Comparison among the four forest change detection parameters

Table 8 provides a summarized comparison among the four forest change detection parameters based on a handful of factors. This table intends to give the readers a clear idea on the variation among the four parameters and also to establish the fact that different remote sensing datasets can yield different results and so type of combination to reach a conclusion is always preferred. The accompanying table presents a thorough examination of the dynamic shifts in vegetation indices (MSAVI, NDVI, and EVI) and land use/land cover (LULC) from 2004 to 2022, along with projections for 2030. The data clearly shows the best and lowest values for every metric, highlighting a general downward trend overall with a noteworthy 2.66 % annual deforestation rate. Moreover, the data emphasizes how disasters affect LULC, MSAVI, and NDVI, highlighting a vulnerability to environmental disturbances. It's interesting to note that EVI shows a declining trend yet is resistant to similar calamities. The rates of change, which show a continuous decrease in both land cover and vegetation health, provide important insights into the transforming dynamics of the ecosystem. The projected values for 2030, which indicate a persistent decline in land cover and vegetation indices, support this tendency. This emphasizes how important it is to keep an eye out and take proactive steps to address environmental issues and stop the degradation from getting worse. The material supplied emphasizes the decreasing trends repeatedly, which highlights the importance and necessity of implementing countermeasures to stop environmental degradation.

Using the three main vegetation indices (NDVI, EVI, and MSAVI), Table 9 presents a succinct summary of the area of healthy vegetation and related trends throughout time. A troubling scenario of deteriorating plant health and probable deforestation is reflected in the overall trend across all three indices, which is a noticeable drop in the extent of healthy vegetation. The NDVI consistently displays the largest area of healthy vegetation, demonstrating its sensitivity to chlorophyll content

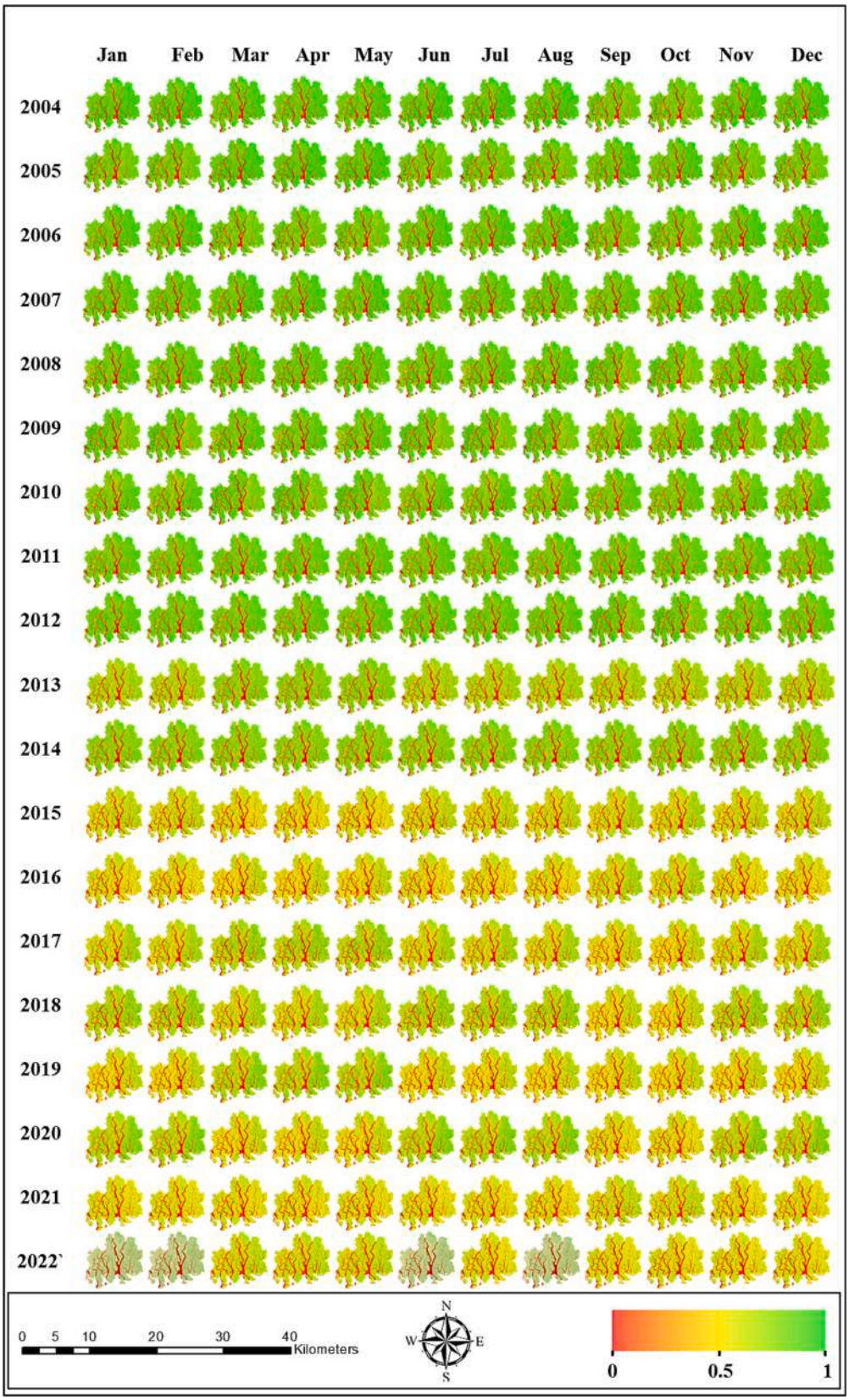


Fig. 10. Index Anatomy of Monthly Spatial Variation of NDVI.

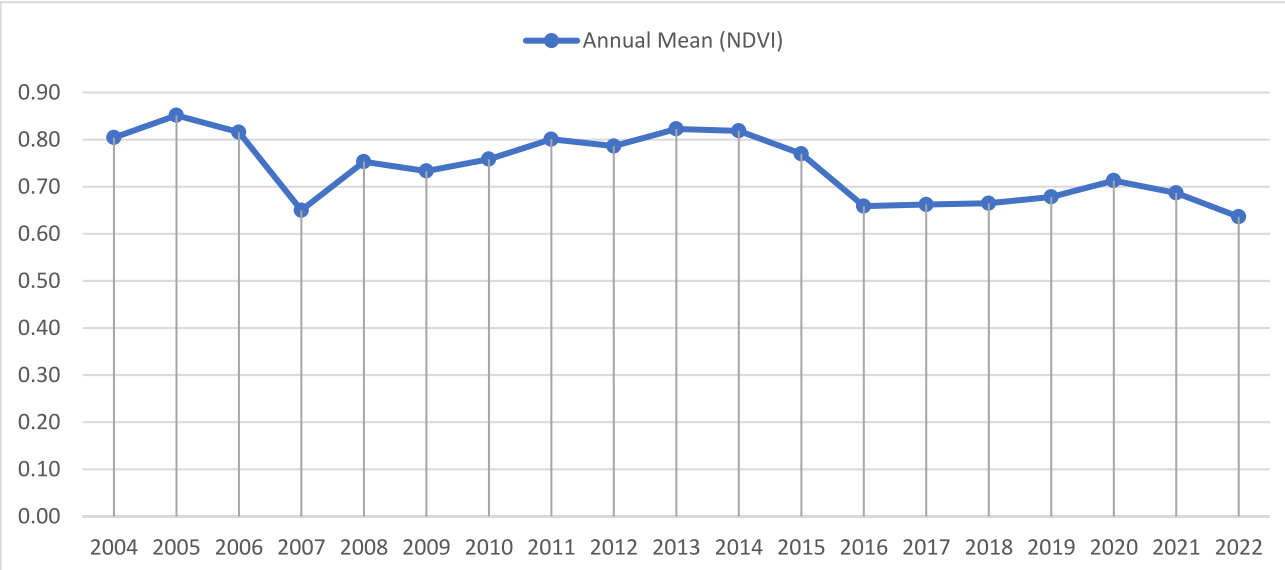


Fig. 11. Annual Time Series of NDVI.

Table 6
Mann-Kendall Trend Test and Sen's Slope for NDVI.

Mann-Kendall Test				
Index	Kendall's Tau	Variance	P-value	Presence of Trend
NDVI	**0.768**	950	0.00352	Yes
Sen's Slopes				
Index	Sen's Slope (Q_i)	Sign of Q_i	Type of trend	Rate of growth
NDVI	**−0.00339**	Negative	Decreasing	−0.00339/year

and capacity to record more significant changes in the health of the vegetation. Conversely, MSAVI consistently displays the least amount of healthy vegetation after adjusting for soil conditions, underscoring its focus on vegetation health regardless of soil impacts. The pattern of declining healthy vegetation that has been seen is consistent with the larger trend analysis and suggests a possible environmental problem. Interestingly, during years when there are natural disasters, the indices as a whole show a consistent decrease in the health of the vegetation. Numerous reasons, such as extreme weather, the effects of climate change, and human activity, might be blamed for this loss because they all contribute to an environment that is less conducive to plant growth (Sarkar et al., 2023). Using NDVI, EVI, and MSAVI simultaneously in this research offers a full view. These are useful methods for monitoring the health of the vegetation. The fact that all three indices show a steadily declining trend highlights the severity of the crisis. The particularly alarming pattern of decline highlights the urgent need for effective environmental conservation measures and sustainable land management techniques. A startling 12.5 % is found when the total rate of decline in healthy vegetation is examined between 2004 and 2022. This percentage highlights the seriousness of the current environmental problem by quantifying the size of the drop. As a quantitative indicator of the urgency of acting to stop deforestation and address the issues

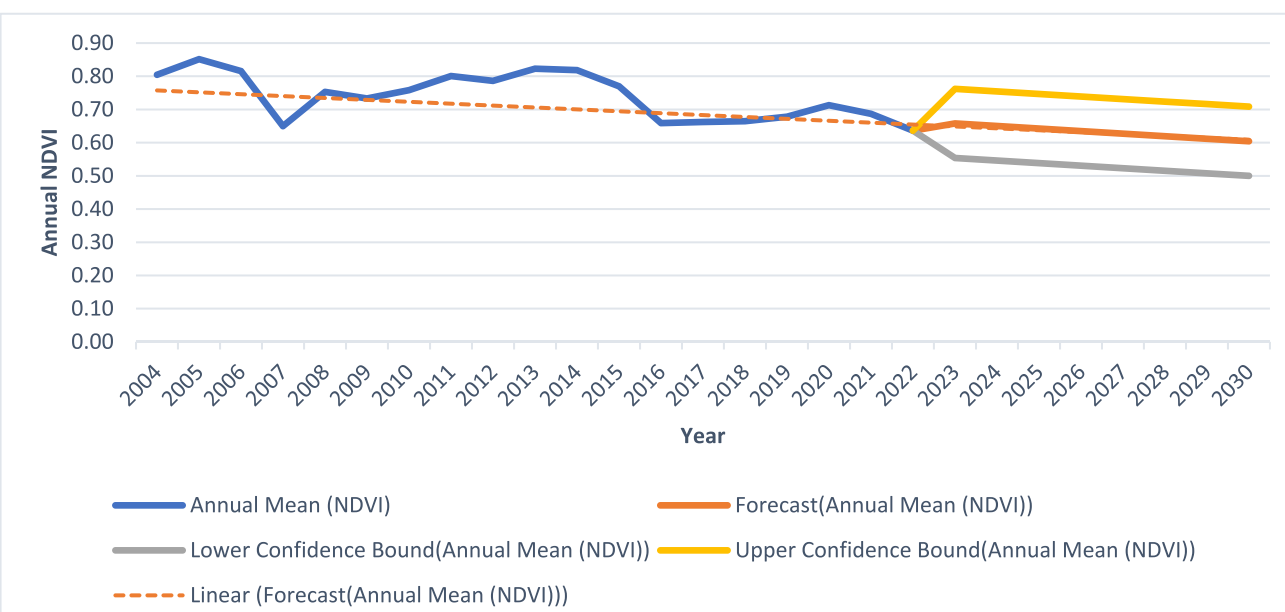


Fig. 12. Forecasting of NDVI using ARIMA. *RMSE: 0.49; MASE: 0.61

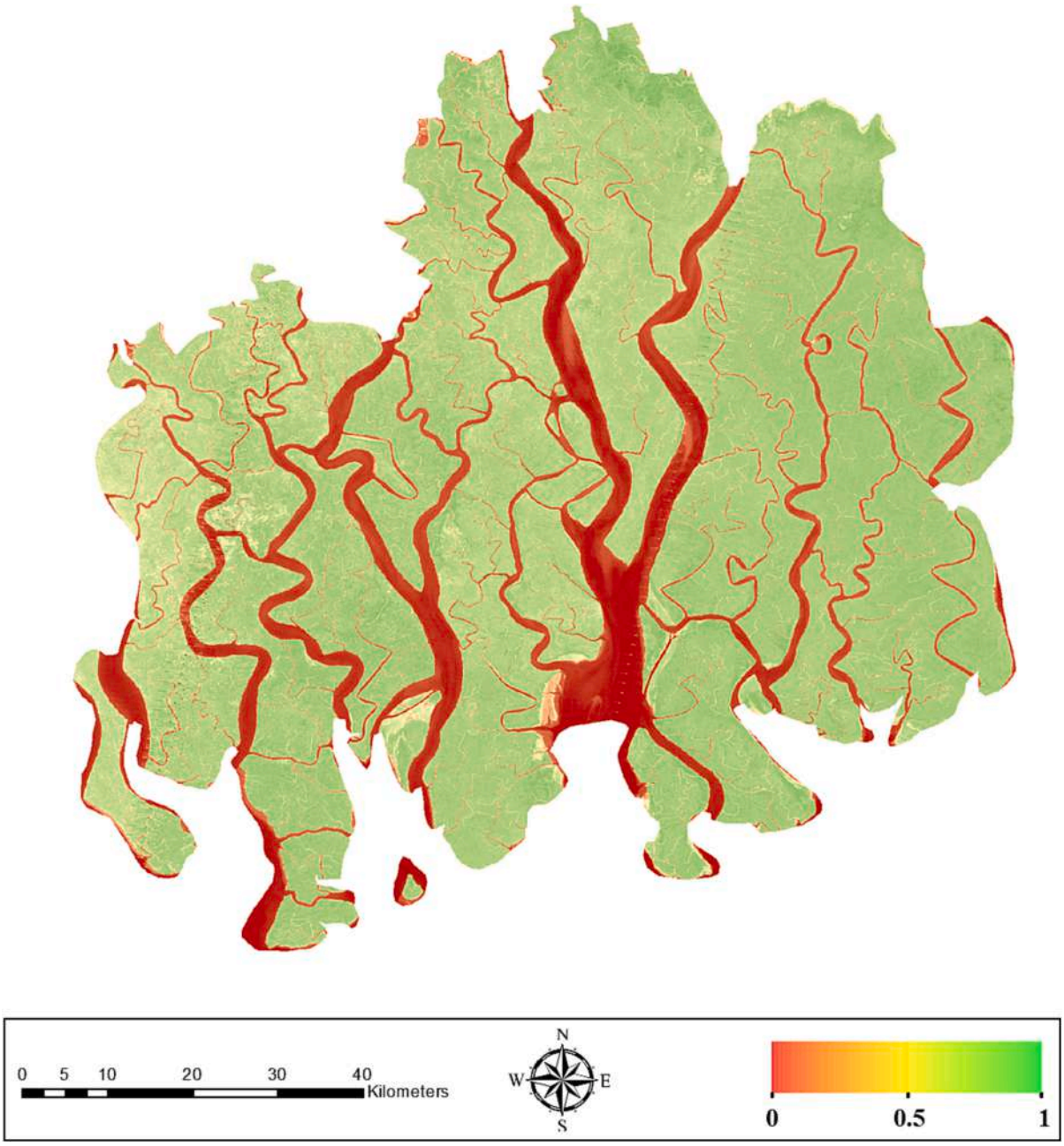


Fig. 13. Spatial Simulation of NDVI for 2030.

causing declining plant health, the rate of reduction is useful. Each index has unique qualities that add subtleties to the analysis. Higher NDVI readings might indicate denser and more resilient plants, but they might also increase variability because of how sensitive they are to atmospheric conditions. EVI, which aims to lessen atmospheric impacts and provide a more accurate assessment of the health of the vegetation, supports the declining trend. The declining trend is consistent with MSAVI, which emphasizes minimizing soil influences and reflects a targeted assessment of vegetation health that is independent of soil conditions. All three indices show a loss in healthy vegetation, which presents serious ecological and environmental risks. Deforestation contributes to climate change, soil erosion, and the disruption of ecosystem services, in addition to the immediate loss of biodiversity and habitat. The obvious effects of natural disasters on the health of plants highlight how susceptible ecosystems are to outside stresses. Finally, a careful examination of Table 9 reveals a concerning trend of deteriorating healthy vegetation over time, according to NDVI, EVI, and MSAVI. Each index's distinct qualities offer insightful information, and the overall pattern points to a serious environmental problem. In order to reduce deforestation and protect plant health, conservation initiatives and sustainable practices should be put into place. The calculated rate of decrease serves as evidence of this, highlighting how serious the problem is. The integrated use of numerous vegetation indices increases the analysis's robustness and also gives a complete picture of the ecological challenges the investigated area faces.

3.6. Annual variation of environmental parameters and Forecasting

Fig 18. shows the Time Series charts of different environmental parameters which also help us gauge the trend and variability of these parameters on an annual scale. Air quality parameters like AOD, NO_2 and CO show the most increasing trend among all the other parameters

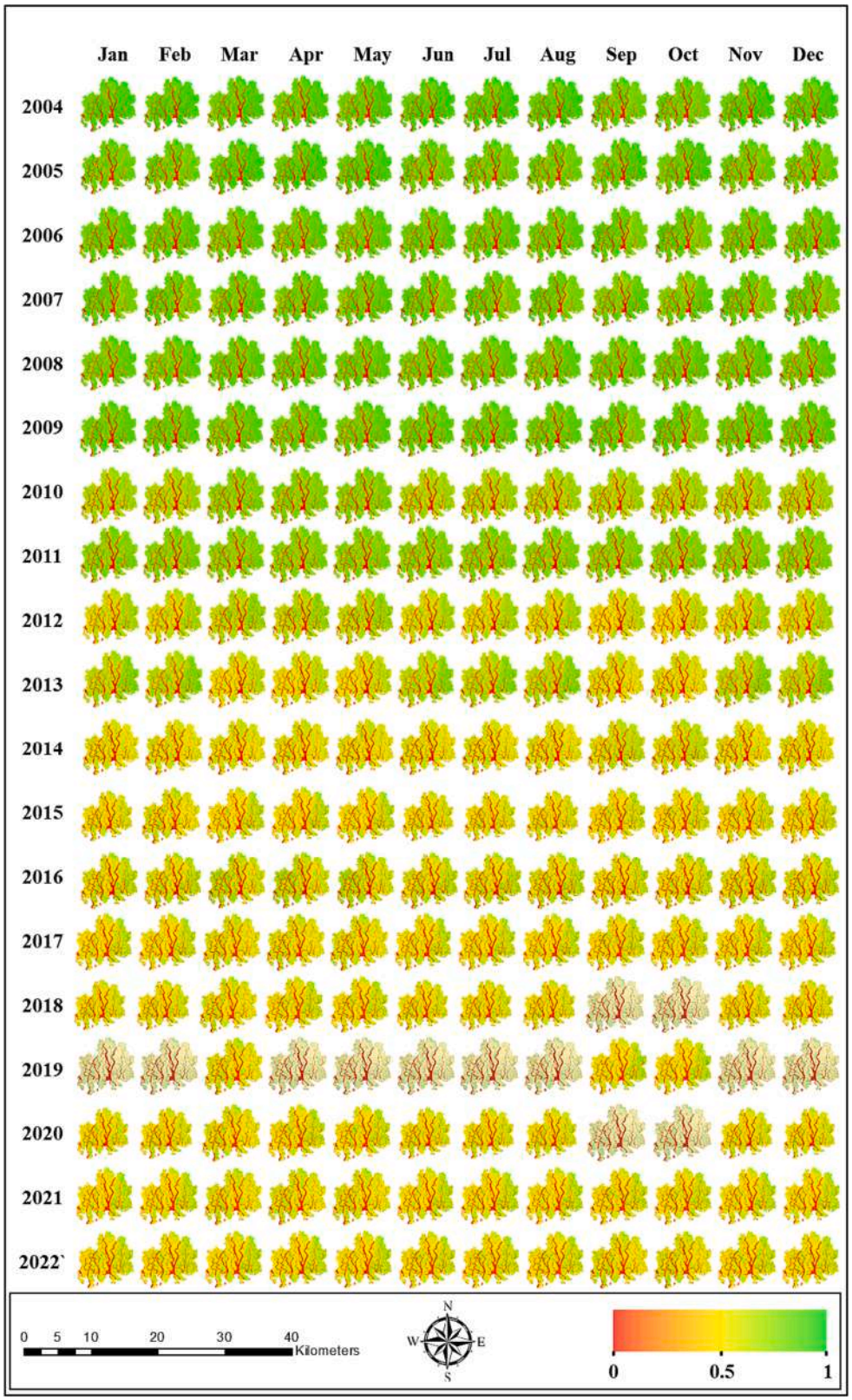


Fig 14. Index Anatomy of Monthly Spatial Variation of EVI.

with an average rate of increase of approximately 8 %, 5.35 % and 1.23 % respectively. This increase in air pollutants particularly is harmful for both wildlife and humans alike who are living around Sundarbans. In contrast to the increasing tendencies observed in AOD, NO2, and CO, the Normalized Difference Salinity Index (NDSI) exhibits little change over time and remains relatively stable. However, the average Land Surface Temperature (LST) peaked in 2023 at roughly 27 degrees Celsius, which is cause for concern as this is the highest LST measured in the last 18 years, suggesting an ongoing warming trend. Causes of elevated LST include urban heat island effects, shifts in land use, and climate change.

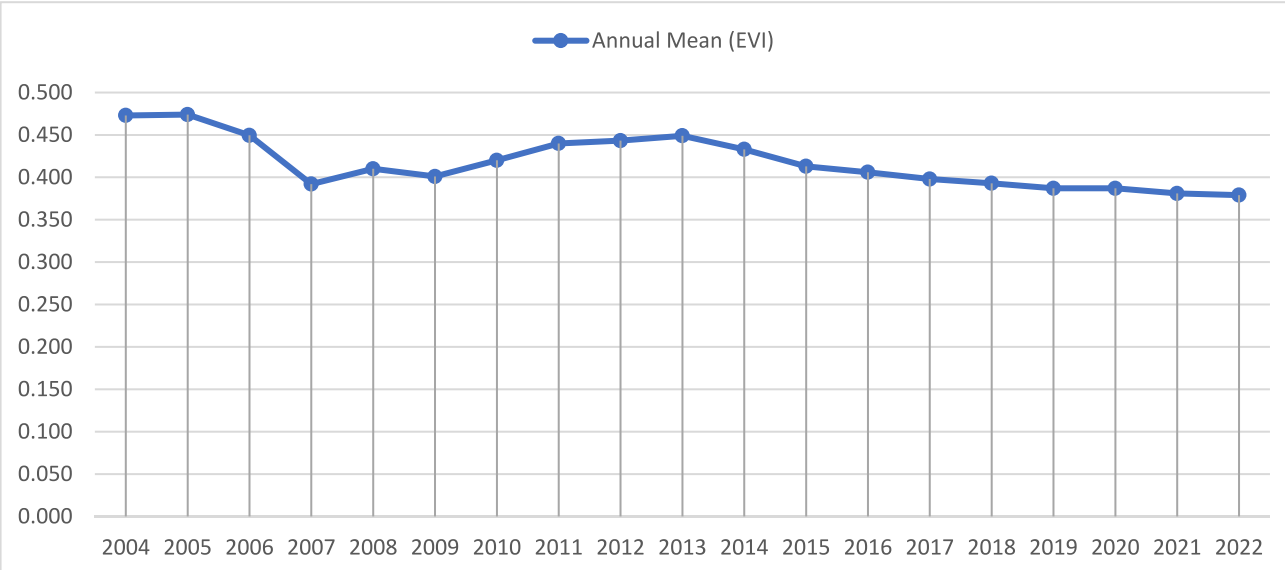


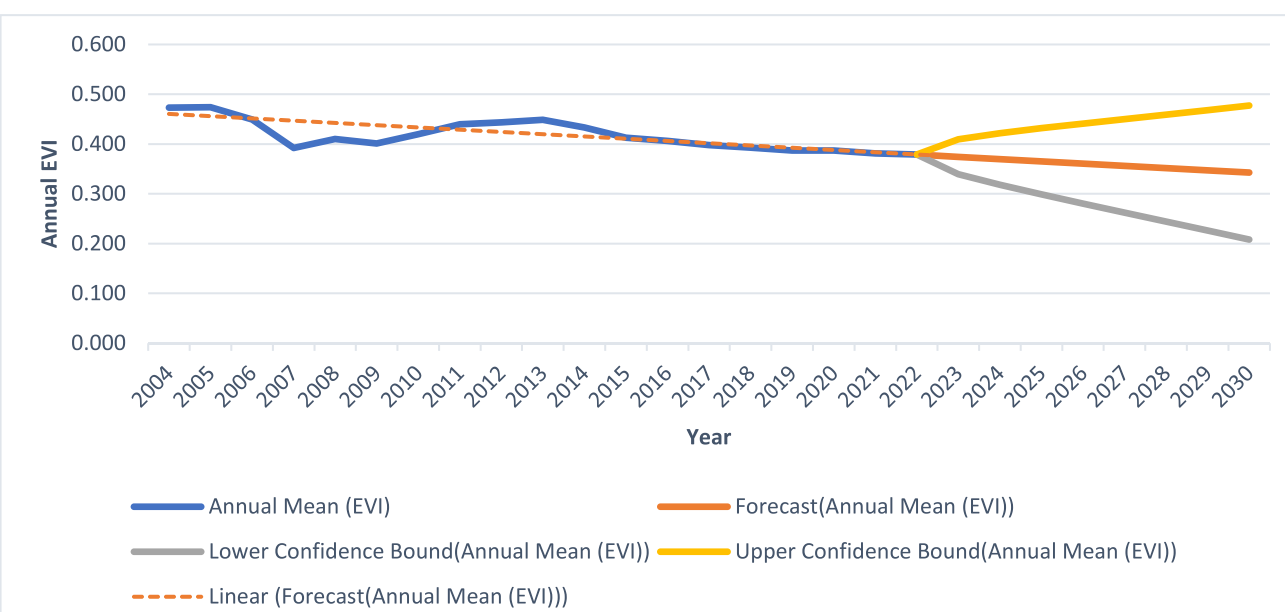
Fig. 15. Annual Time Series of EVI.

Table 7
Mann-Kendall Trend Test and Sen's Slope for EVI.

Mann-Kendall Test				
Index	Kendall's Tau	Variance	P-value	Presence of Trend
EVI	**0.635**	950	0.00023	Yes
Sen's Slopes				
Index	Sen's Slope (Q_i)	Sign of Q_i	Type of trend	Rate of growth
EVI	**−0.00504**	Negative	Decreasing	−0.00504/year

Climate Water Deficit which refers to the situation when environmental needs (for plants, animals, and ecosystems) exceed the availability of water (from precipitation and soil moisture) also shows an increasing trend. This trend can be exacerbated by climate change, changing precipitation patterns, and land-use changes, creating threats to water supply, agricultural productivity, and ecological balance (Das and Sarkar, 2023). Inversely, Soil Heat Flux which is the rate of heat exchange between the soil and the surrounding environment, shows a decreasing trend along with Soil Moisture and Precipitation which is decreasing at the rate of approximately 2.1 % and 1.9 % respectively. Changes in land cover, vegetation dynamics, or the energy balance as a result of urbanization are all possible explanations for this drop.

Fig. 19 shows environmental conditions, and the anticipated environmental parameters shed light on shifting trends and potential causes for concern. The forecast highlights rising air pollution as a major concern, with annual mean values for AOD (aerosol optical depth) increasing by 60 % by 2030 compared to the base year of 2004, NO2 increasing by 19 %, and CO increasing by 22 %. These dramatic increases are indicative of deteriorating air quality, especially in forested regions, and may pose a risk to the ecological system as a whole. On the other hand, after an abnormal increase in 2022, the trends of land



*RMSE: 0.39; MASE: 0.86;

Fig. 16. Forecasting of EVI using ARIMA. *RMSE: 0.39; MASE: 0.86;

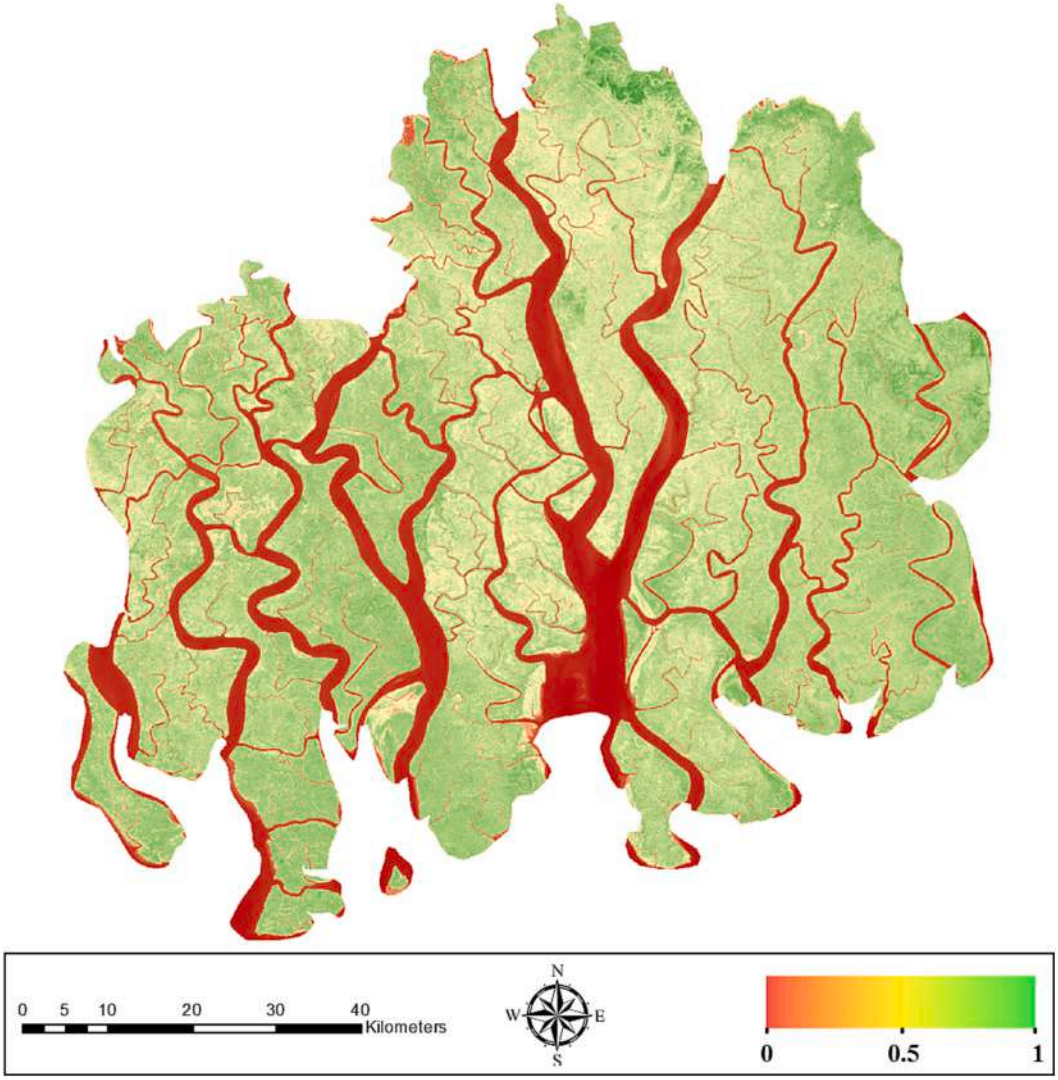


Fig. 17. Spatial Simulation of EVI for 2030.

Table 8
Comparison among four forest change detection parameters.

Criteria	LULC	MSAVI	NDVI	EVI
Highest Year	2004	2005	2005	2005
Lowest Year	2022	2007	2022	2022
Highest Value	3531.16 Sq. km	0.188	0.85	0.474
Lowest Value	3065.80 Sq. km	0.125	0.64	0.379
Influenced by Disasters	Yes	Yes	Yes	No
Presence of Trend	Yes	Yes	Yes	Yes
Type of Trend	Decreasing	Decreasing	Decreasing	Decreasing
Rate of deforestation/year	2.66 %	0.00425 %	0.00339 %	0.00504 %
Forecasted Value in 2030	2846.97 Sq. km	0.123	0.60	0.343

surface temperature (LST) and normalized difference salinity index (NDSI) are rather constant. This steadiness suggests that certain environmental influences may not require immediate attention. But it's important to keep an eye on these numbers to make sure they stay steady over time. An alarming 60 % increase from 2004's baseline is expected for climate water deficit, a key indicator of water scarcity. Water scarcity has the potential to have devastating effects on ecosystems, agriculture,

Table 9
Annual variation of healthy vegetation area according to three employed indices.

Year	NDVI (in sq.km)	EVI (in sq.km)	MSAVI (in sq.km)
2004	3178.044	3036.7976	2895.5512
2005	3056.94	2921.076	2785.212
2006	3124.638	2985.7652	2846.8924
2007	2788.317	2664.3918	2540.4666
2008	2974.203	2842.0162	2709.8294
2009	3055.68	2919.872	2784.064
2010	3029.022	2894.3988	2759.7756
2011	3122.073	2983.3142	2844.5554
2012	3095.325	2957.755	2820.185
2013	2883.69	2755.526	2627.362
2014	2945.052	2814.1608	2683.2696
2015	3025.557	2891.0878	2756.6186
2016	2938.635	2808.029	2677.423
2017	2958.696	2827.1984	2695.7008
2018	3018.483	2884.3282	2750.1734
2019	2825.631	2700.0474	2574.4638
2020	2868.84	2741.336	2613.832
2021	2878.542	2750.6068	2622.6716
2022	2759.22	2636.588	2513.956

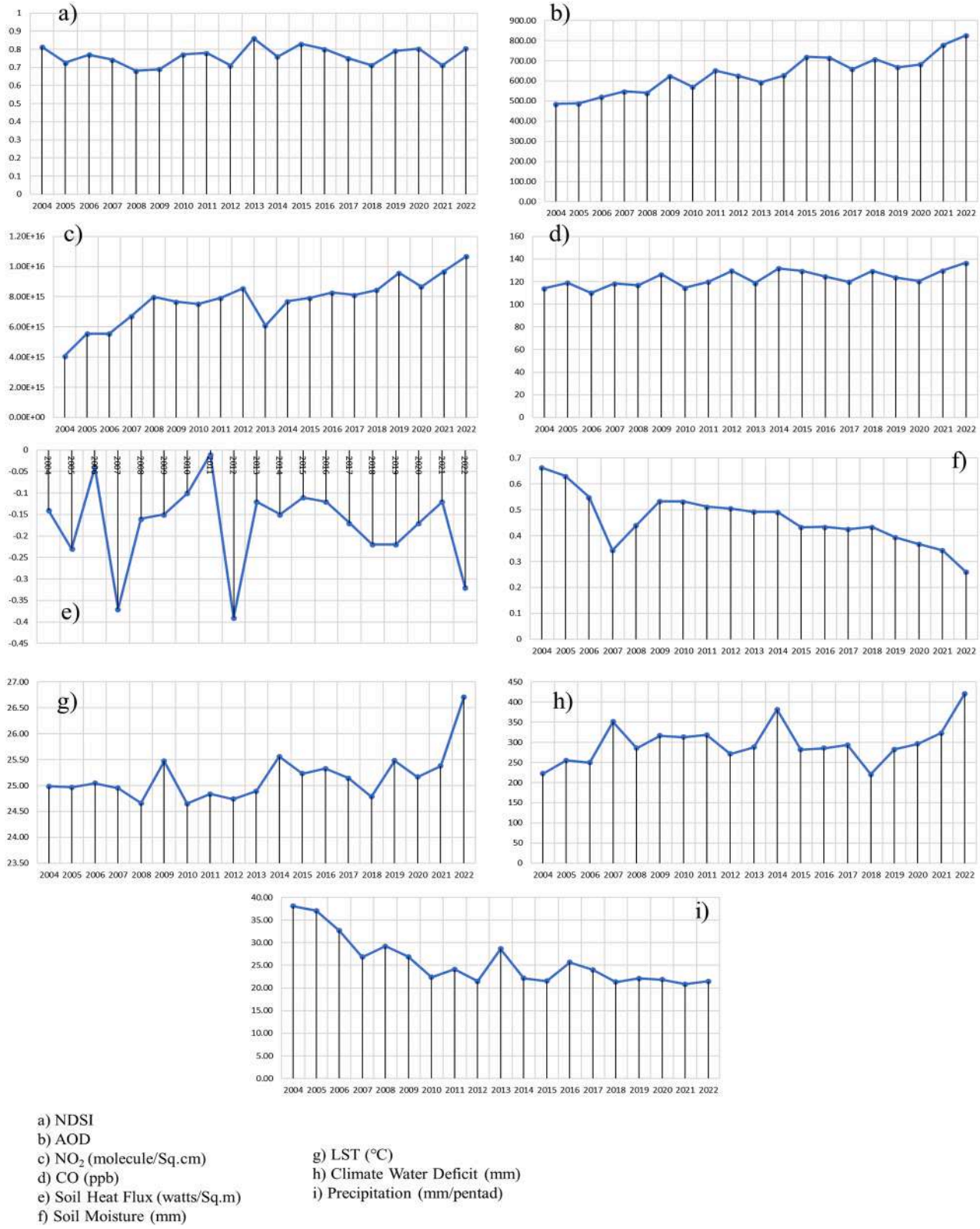


Fig. 18. Annual Variation of Environmental Parameters using Time-Series.

and human populations in the areas where it occurs. The expected increase will have negative consequences that must be mitigated through efficient water management systems and conservation initiatives. There is no clear pattern in the forecast for soil heat flux, suggesting that no major shifts in this parameter are to be anticipated for the time period being considered. Soil heat flux plays a crucial role in the energy exchange between the land surface and the atmosphere, although more research may be needed to fully comprehend its significance. Finally, the projection calls for a 39 % drop in precipitation and a 23 % drop in soil moisture from their 2004 highs. Drought conditions and the availability

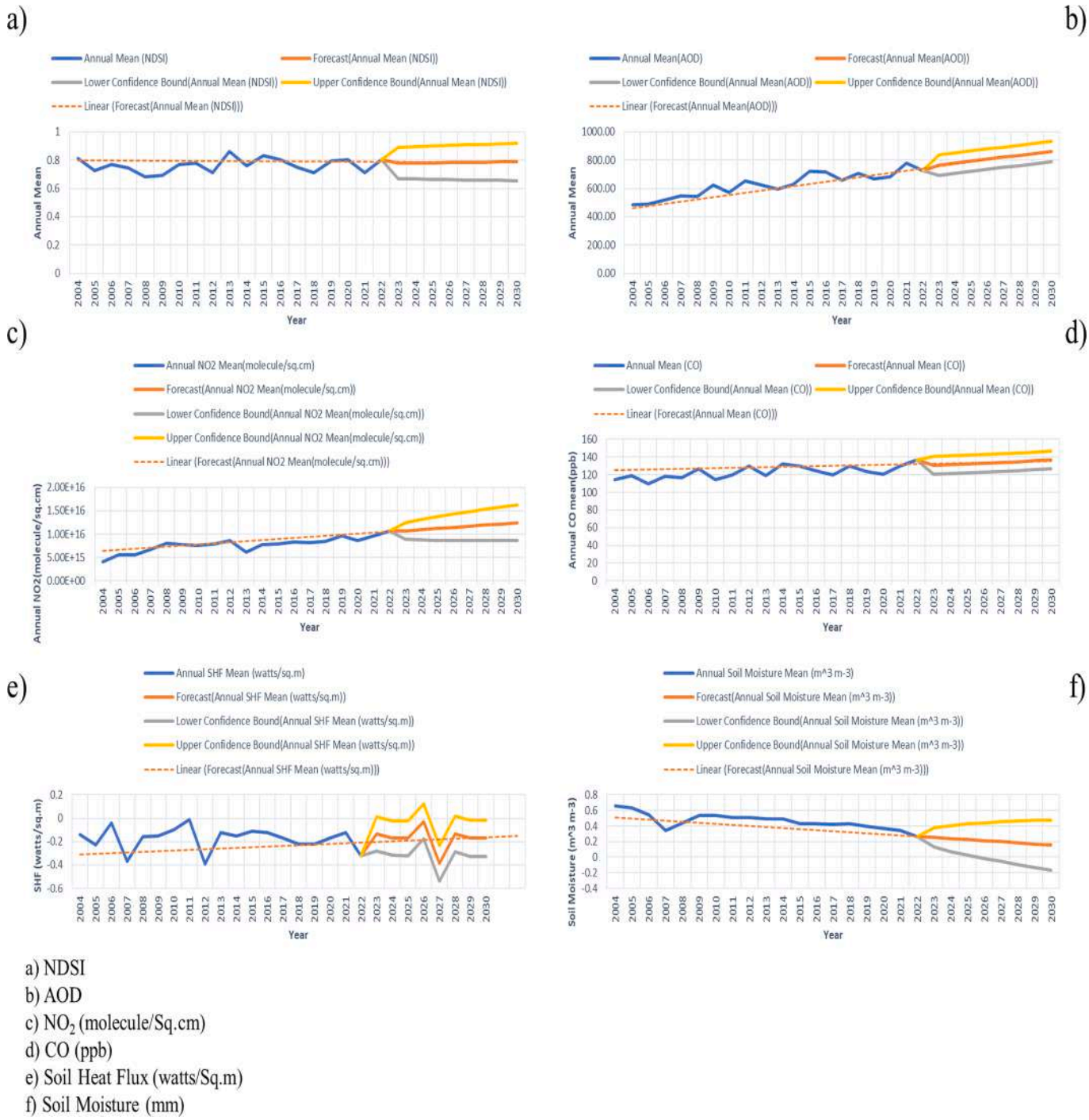


Fig. 19. Forecasting of Environmental Parameters using ARIMA. *RMSE: 0.36; MASE: 0.53

of water supplies become causes for concern due to these cuts, which can have far-reaching effects on ecosystems, agriculture, and water security as a whole.

The predicted changes in environmental factors highlight a number of serious issues that represent serious threats to human health, water supply, and ecosystems. An alarming situation is created by the growth of air pollution, the sharp rise in the climate water deficit, declining precipitation and soil moisture levels, and other factors. The effects on ecosystems are profound because these modifications have the potential to upset sensitive ecological balances, jeopardizing ecosystem services and biodiversity. The availability of water supplies for agriculture and human consumption is threatened by the growing water deficit and changing climatic trends. The problems are further made worse by decreasing soil moisture and precipitation, which can result in drought conditions and lower agricultural yield. The increased risk of respiratory illnesses and other health problems is one of the major hazards to human health posed by the elevated levels of air pollution. Proactive actions are necessary to meet these upcoming difficulties. It is imperative to implement policies like strict pollution control, sustainable water management, and coordinated efforts to lessen the wider effects of climate change. In order to ensure a healthier and more secure future for future generations, it is imperative that preventative activities be taken

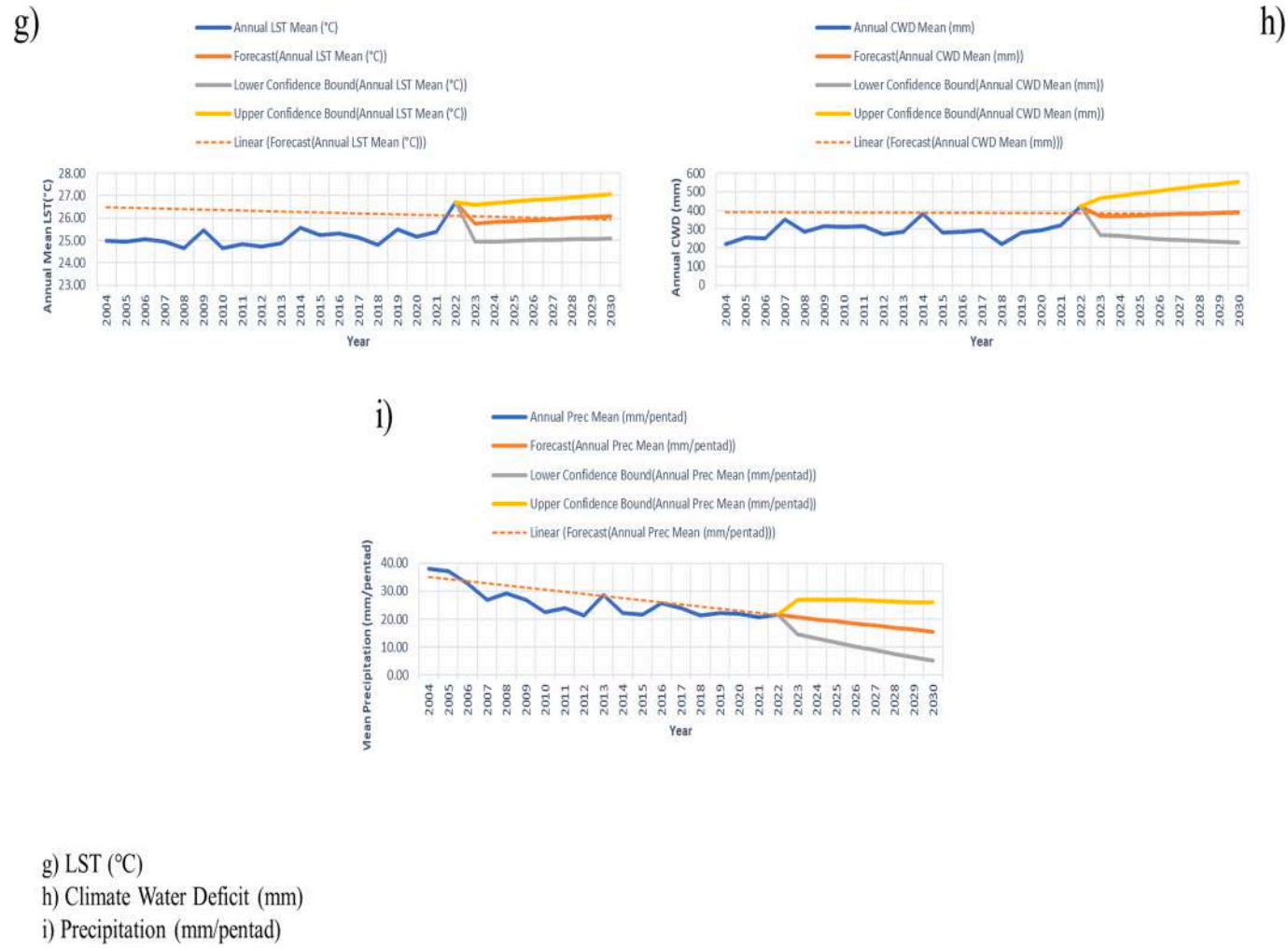


Fig. 19. (*continued*).

in order to create a resilient and sustainable environment that can resist the difficulties presented by changing environmental conditions.

3.7. Correlation among forest change dynamics and environmental parameters

Auto-Correlation was performed to decipher the nature of correlation by utilizing the correlation coefficients shown in Table 10, while lagged correlation was performed to gauge the tendency of past values of forest change parameters to influence present values of environmental factors shown in Table 11..

Notwithstanding the degree of correlation, the correlation coefficients show that LULC and NDVI have somewhat comparable effects on environmental variables. A rise in NDSI, AOD, CO, NO2, LST, and CWD is correlated with a decrease in precipitation, soil moisture, soil heat flux, and soil moisture content, and a drop in NDVI or forest area (LULC). Remarkably, a drop in NDVI has a strong impact on AOD and Soil Moisture, with NO2 and precipitation following closely behind. In contrast, LULC has a greater effect on CO and soil moisture, with a smaller effect on precipitation and soil heat flux. The decline in EVI and MSAVI also leads to comparable correlations; the loss in EVI is correlated with a rise in SHF, while the decline in MSAVI helps to reduce the Climate Water Deficit. Changes in MSAVI mostly impact AOD and CO, with NO2 and precipitation following closely behind, while EVI has the greatest impact on Climate Water Deficit and LST. In order to make well-informed decisions, it is crucial to consider a variety of remote sensing indices and factors, as demonstrated by the varied connections between numerous forest change metrics and environmental conditions. The observed patterns in the effects of forest change on the environment highlight how intricate these linkages are.

The interplay between environmental conditions and forest transformation is nuanced and can have both short- and long-term effects. Depending on the lag time between the onset of one environmental condition and its effect on forest dynamics, the relationship between these elements can be shown to be either linear or nonlinear (Chakraborty et al., 2022). The relationships between forests and the

Table 10
Auto-Correlation coefficients.

Index	NDSI	AOD	CO	LST	NO2	CWD	Precipitation	SHF	SM
LULC	−0.23	−0.33	−0.49	−0.12	−0.36	−0.25	0.67	0.49	0.45
NDVI	−0.11	−0.60	−0.37	−0.15	−0.54	−0.29	0.39	0.38	0.57
MSAVI	−0.006	−0.45	−0.47	−0.07	−0.35	0.18	0.35	0.25	0.03
EVI	−0.17	−0.11	−0.19	−0.28	−0.05	−0.54	0.007	−0.11	0.26

Table 11
Cross-correlation coefficients of Forest change parameters against environmental parameters with lags in years.

Index	Factors	Lags in years from base year 2004					
		0	1	2	3	4	5
LULC	**NDSI**	−0.23	−0.34	0.41	0.21	−0.53	−0.49
	AOD	0.52	−0.25	0.32	−0.44	0.42	0.24
	CO	−0.33	−0.45	0.36	−0.22	−0.43	0.56
	LST	0.26	0.44	−0.67	−0.45	−0.33	0.31
	NO2	−0.08	−0.005	0.33	0.44	−0.56	−0.87
	CWD	−0.16	−0.12	0.44	0.78	−0.33	−0.45
	PREC	0.21	0.45	0.02	−0.46	−0.78	−0.56
	SHF	0.31	−0.03	−0.06	0.05	−0.56	0.01
	SM	0.07	0.54	−0.26	−0.39	−0.37	0.31
NDVI	**NDSI**	−0.27	−0.24	0.21	0.17	0.11	−0.06
	AOD	0.15	−0.18	0.33	−0.32	0.60	0.59
	CO	0.72	−0.41	−0.28	−0.11	0.37	0.39
	LST	0.22	−0.17	0.20	−0.14	−0.10	0.12
	NO2	0.08	0.86	−0.28	−0.38	0.54	−0.36
	CWD	−0.17	−0.12	0.05	0.27	0.29	−0.22
	PREC	0.20	−0.08	0.14	−0.13	0.39	−0.19
	SHF	0.01	0.03	−0.02	0.07	−0.37	−0.21
	SM	−0.09	0.04	0.21	−0.49	−0.57	0.55
MSAVI	**NDSI**	0.38	−0.28	0.63	−0.03	−0.06	0.14
	AOD	−0.17	−0.22	0.26	−0.17	0.45	0.25
	CO	0.41	0.03	−0.21	−0.22	0.47	−0.17
	LST	0.37	0.55	0.92	−0.14	−0.06	−0.34
	NO2	0.74	−0.17	−0.15	0.27	0.35	−0.21
	CWD	0.33	0.10	0.21	−0.07	−0.18	−0.12
	PREC	−0.12	0.39	−0.13	0.04	−0.35	0.02
	SHF	0.01	−0.20	−0.31	−0.05	0.25	0.04
	SM	−0.14	0.22	0.16	0.05	−0.03	−0.04
EVI	**NDSI**	−0.12	0.13	−0.05	−0.09	0.16	0.39
	AOD	−0.21	−0.02	0.29	0.28	−0.10	0.16
	CO	0.15	−0.01	0.44	−0.07	−0.19	0.02
	LST	0.27	−0.24	0.37	0.02	−0.27	−0.28
	NO2	0.15	−0.15	0.05	0.20	−0.05	−0.11
	CWD	0.18	0.03	−0.10	−0.54	−0.25	0.11
	PREC	0.24	−0.15	−0.57	−0.02	0.10	0.18
	SHF	0.26	−0.02	−0.57	−0.02	0.10	0.18
	SM	−0.16	0.43	0.45	0.35	−0.26	−0.05

environment can be made simpler by identifying these patterns. To clarify, let us utilize 2004 as the starting year, which we will represent as 0. Then, ascending numbers are used to indicate the following years: 1 for 2005, 2 for 2006, and so on. When environmental factors and forest dynamics are examined together, it becomes clear that some things affect forests immediately while others have consequences that take time to manifest. AOD, LST, CWD, and CO levels exhibit immediate responses to forest changes, suggesting a close relationship between these dynamics and observed environmental changes. On the other hand, some environmental elements respond slowly; because forest dynamics play a role, it takes about a year for noticeable effects to manifest. A more complex interconnectedness, potentially involving intermediary processes that mediate observable changes in the forest ecosystem, is suggested by this delayed effect. The temporal lag highlights how intricately environmental elements and forest transformation interact. The resilience of the ecosystem is gradually undermined by ongoing forest alterations until it inevitably fails. Significantly, changes and trends become more noticeable across all four forest change parameters starting at the third lag. This suggests that while the ecosystem may initially be able to tolerate the consequences of forest change, more drastic or protracted changes may eventually have an adverse effect.

4. Discussion

The use of remote sensing datasets has undergone a revolutionary development, expanding in availability and accessibility beyond all previous limits. According to (Mondal and Debnath, 2017), this change not only indicates that data is now more widely available, but it also highlights how inexpensive and simple it is to access when compared to more conventional methods. The benefits of remote sensing, in terms of cost and accessibility, make it an effective tool for environmental monitoring even though efforts to maintain data consistency are still needed. Time series research on large datasets—often referred to as big Earth data—offers a singular chance in the field of environmental studies to identify trends and correlations over geography and time (Rahman, 2012). This study takes advantage of this chance to examine the dynamics of forest change on the ecosystem by concentrating on the Sundarbans, the world's biggest continuous mangrove forest. Land Use and Land Cover (LULC), Normalized Difference Vegetation Index (NDVI), Modified Soil-Adjusted Vegetation Index (MSAVI), and Enhanced Vegetation Index (EVI) are the four forest change variables that are being examined. Using these variables, the study spans the years 2004 to 2022 and aims to decipher the complex fluctuations that occur in the Sundarbans and to determine how these variations affect a variety of environmental parameters, such as NDSI, AOD, CO, NO2, Land Surface Temperature (LST), Climate Water Deficit, Precipitation, Soil Heat Flux, and Soil Moisture. The study's conclusions illuminated the Sundarbans' continuous deforestation and reduction of forest cover, showing a concerning rising trend. Known as Bangladesh's shield, the Sundarbans are the biggest carbon sink, reducing the damaging impacts of high temperatures and greenhouse gasses. They also play a critical role in defending Bangladesh against natural calamities(Mondal and Debnath, 2017). The study's findings about the decline in forest cover have significant effects on the populace. It affects socioeconomic circumstances and the means of subsistence for those reliant on the Sundarbans, in addition to environmental issues(Rahman et al., 2010). The study's forest change variables draw attention to the steadily declining forest acreage and vegetation, which presents serious problems. According to corresponding forecasting and simulation models, there will be a significant change in the upcoming years, with an average annual decline in forest area of 2.66 %. The LULC simulation, which projects a startling 600 sq. km drop in forest area by 2030, is especially concerning. The various issues encountered by the Sundarbans are highlighted by this projection, which is linked to causes including infrastructural development, fast expansion of human habitat, and forest cutting. The research highlights rising air pollution and surface temperatures, as well as falling soil moisture and precipitation, as some of the significant effects of this declining forest cover on the environment. According to (Chongling et al., 2009), these modifications may have both immediate and long-term repercussions on the ecosystems of the forests, further reducing the health of the forest vegetation and having more detrimental effects on the environment. Significant interactions and effects of the forest on the environment are revealed by the cross-correlational studies carried out in this research. For the first year or two, the ecosystem might be more resilient, but eventually it gives in to the never-ending cycle of deforestation and environmental degradation. The Sundarbans require immediate implementation of ecologically appropriate land-use planning, conservation, and management strategies due to this cyclical trend (Thakur et al., 2021). In order to preserve the Sundarbans, the report promotes teamwork among the government, locals, scholars, and environmental organizations. It is believed to be imperative to implement early interventions and adaptable management techniques that are based on monitoring changes in vegetation health, air quality, and forest cover. The Sundarbans are vital to the region's environmental sustainability because they provide a shelter from storms and preserve the livelihoods of millions of people (Sahana et al., 2015). Given the Sundarbans' ecological significance, the study highlights the need of proactive conservation measures to guarantee this priceless ecosystem's survival for future generations (Shimu et al., 2019). The study does, however, agree that more inquiry and study are necessary to precisely identify the sources, reasons, and wider ramifications of the observed inclinations. Additionally, because of data

unavailability, the study emphasizes the limits imposed by the lack of some environmental characteristics in its analysis. This disparity emphasizes how important it is to improve data availability and collecting in order to obtain more detailed understanding of the intricate dynamics influencing the Sundarbans. As a result, this report offers a critical analysis of the environmental issues the Sundarbans are facing and issues a call to action for sustainable development and conservation. A thorough grasp of the current changes is provided by the integration of remote sensing, time series analysis, and forecasting models, highlighting the necessity of concerted and well-informed efforts to preserve and restore this priceless ecosystem.

5. Conclusion

In conclusion, this research examines the effects of recent forest-related changes in the Sundarbans region on the region's biodiversity and ecosystem health. The results show that deforestation and land conversion for agricultural and infrastructure development have led to a dramatic decrease in forest cover over the previous decade. The unique and diversified Sundarbans ecology is threatened by the clearance of forest land. The natural equilibrium and ecological functions of the area are disturbed by the increase in developed land in the region, which adds to the decline in forest cover. The study also shows that the vitality and health of Sundarbans plants are declining. The frequency and severity of these forest change or decline has caused serious nefarious impacts on the environment as well as the forest wildlife and ecology. These shifts have far-reaching consequences as increase of air pollutants, temperature, decrease in precipitation and soil moisture and negative variations of other parameters hampers not only the forests health and but also the lives of the population and the planet as a whole (Ali et al., 2021). The Sundarbans not only serves as a lifeline for many plant and animal species, but also for the people who live in the surrounding areas (Uddin et al., 2013). Mangrove trees defend coastlines from storms and erosion, keep people safe from flooding, and provide food and income through fishing and tourism and communities are made more vulnerable to natural disasters and deprived of critical resources when forest cover and vegetation health are destroyed (Mukul et al., 2020). The Sundarbans' preservation is crucial not just for environmental reasons, but also for social and economic ones. It is essential to educate locals about the Sundarbans' worth and incorporate them into preservation initiatives. We can maintain the Sundarbans for future generations and the well-being of the region's wildlife and human communities if we acknowledge the ecosystem's importance and take proactive actions to preserve and restore it.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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